

CONSTRUCTION SPECIFICATIONS

**SPECIFICATIONS
FOR THE
CONSTRUCTION OF**

**NEW WOOD FLOORING AND
TELESCOPING BLEACHERS
FOR
JOHNSONVILLE HIGH SCHOOL
GYMNASIUM
JOHNSONVILLE, SC**

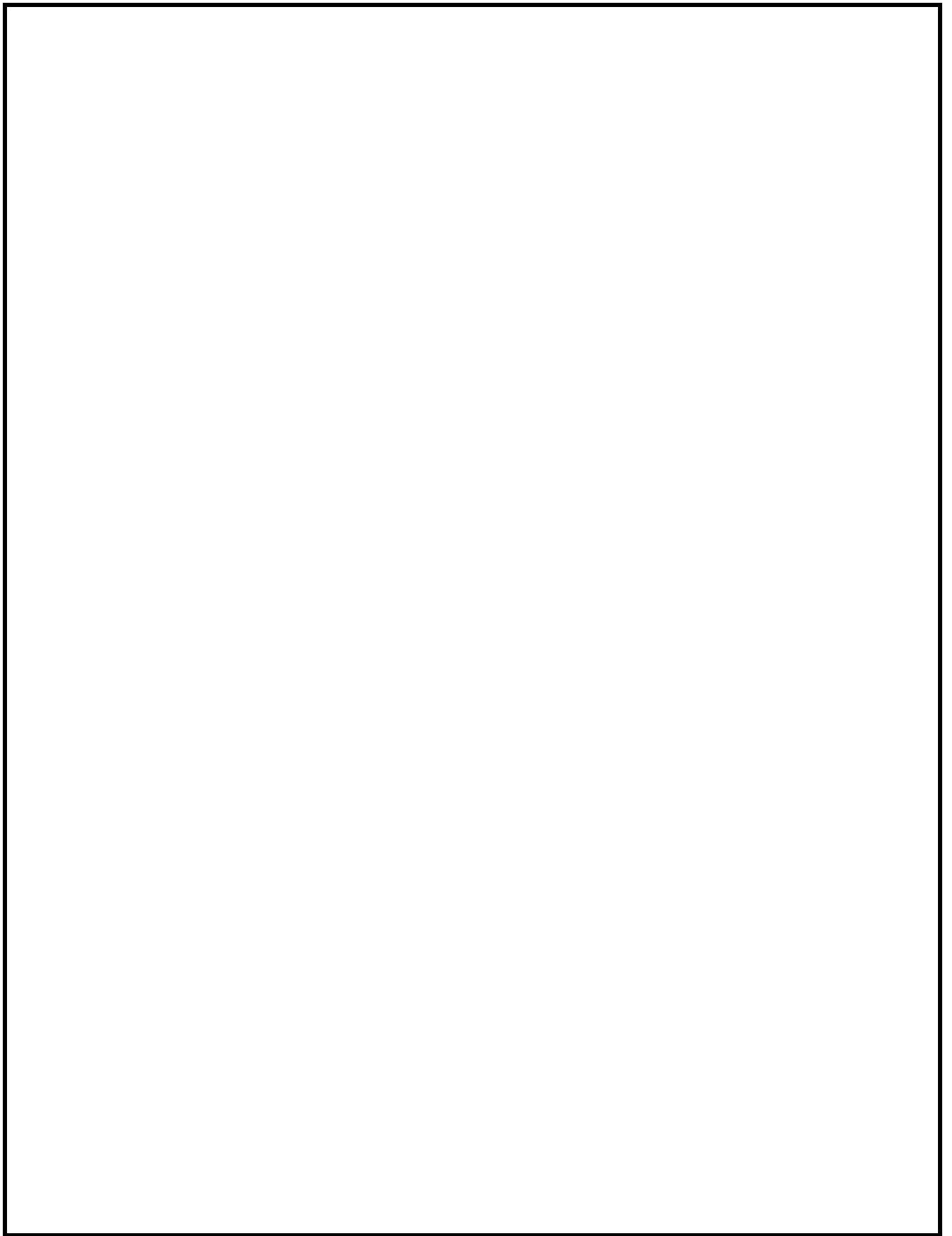
**FLORENCE SCHOOL DISTRICT FIVE
JOHNSONVILLE, SOUTH CAROLINA**

FW ARCHITECT'S COMM. NO. 2303



FW ARCHITECTS, INC. A.I.A.
ARCHITECTURE • PLANNING • INTERIORS
P.O. BOX 2261 FLORENCE, SOUTH CAROLINA

SET NUMBER _____



SPECIFICATIONS

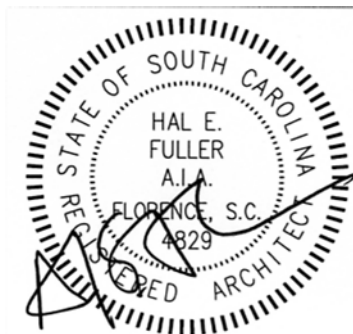
**DATED
FEBRUARY 6, 2023**

**ARCHITECT'S COMM. NO.
2303**

**SPECIFICATIONS
FOR THE
CONSTRUCTION OF
NEW WOOD FLOORING AND
TELESCOPING BLEACHERS
FOR
JOHNSONVILLE HIGH SCHOOL GYMNASIUM**

**FLORENCE SCHOOL DISTRICT FIVE
JOHNSONVILLE, SOUTH CAROLINA**

FW ARCHITECT'S COMM. NO. 2303



**NOTE: TO GENERAL CONTRACTORS, SUBCONTRACTORS AND ALL MATERIAL SUPPLIERS:
THERE SHALL BE NO ASBESTOS CONTAINING MATERIAL USED IN THE CONSTRUCTION OF THIS
PROJECT IN ANY FORM WHATSOEVER.**

**NOTE: ALL CONSTRUCTION MUST MEET ALL SEISMIC REQUIREMENTS OF THE 2021
INTERNATIONAL BUILDING CODE – LATEST APPROVED VERSION.**

**F W ARCHITECTS, INC., AIA
1550 WEST EVANS STREET
FLORENCE, SOUTH CAROLINA 29501**

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JOHNSONVILLE, SC

FLORENCE SCHOOL DISTRICT FIVE
JOHNSONVILLE, SOUTH CAROLINA**

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DOCUMENT 00 11 16 - INVITATION FOR BIDS

Sealed bids for the **INSTALLATION OF NEW WOOD FLOORING AND TELESCOPING BLEACHERS** from qualified bidders, properly licensed under the laws of the State of South Carolina, will be opened and read aloud by the Owner and Architect at the office of F W Architects, Inc., located at 1550 West Evans Street, Florence, SC on March 2, 2023 . Bids received after 2:00 pm will be rejected and returned to the Bidder unopened .

Project Name: **INSTALLATION OF NEW WOOD FLOORING AND TELESCOPING BLEACHERS FOR JOHNSONVILLE HIGH SCHOOL, JOHNSONVILLE, SC**

Pre-Bid Date & Time: N/A BUT BIDDERS ARE ENCOURAGED TO VISIT THE SITE PRIOR TO SUBMITTING A BID.

Bid Date & Time: **March 2, 2023 @ 2:00 p.m.**

Bid Opening Location **F W Architects, Inc. Office**
1550 West Evans Street
Florence SC 29501

Project Address: **Johnsonville High School**
237 S. Georgetown Highway
Johnsonville, SC 29555
843 386-2707

The Owner is: **Florence School District Five**
156 East Marion Street
Johnsonville, SC 29555
843 386-2358

The Architect/Engineer is: **F W Architects, Inc.**
1550 West Evans Street
Florence, South Carolina 29501
Telephone: (843) 662.9961
Fax: (843) 665.5065
Contact: Hal Fuller
Hfuller@fw-architects.com

The scope of work is: Removal of the existing poured urethane floor to the concrete deck. Correct in damaged areas, install a 6 mil vapor barrier over slab and install new wood flooring system as specified. Provide new doors and frames at openings where raised floor is being installed. Provide ramps and landings as shown on drawings. Install electrically operated telescoping bleachers with integral solid plastic seats.

Alternate #1: Price for painting exposed ceiling area with dry-fall paint.

Time of Completion: 60 days. Work to start June 5, 2023 and be completed by August 5, 2023

Bidding Documents will be available for inspection at the offices of the Architect. Bid Documents will be Drop Boxed to qualified bidders at no charge or a CD can be purchased for the non-refundable price of \$25.00 per CD. Checks should be made payable to FW Architects, Inc.

All bids are to be accompanied by a bid bond or certified check of not less than five percent (5%) of the base bid. Bid securities of the three low bidders will be retained until Contract award and execution.

Bids will be binding for a period of thirty (30) calendar days from the date bids are opened.

The successful bidder will be required to furnish Performance Bond and Labor and Materials Bond in the amount of one hundred percent (100%) of the Contract Amount.

Bidders' attention is directed to the requirements of the construction schedule, which is a part of the bidding documents and to the liquidated damages conditions outlined in the Invitation for Bid.

Bids are to be submitted on the bid form provided, enclosed in a sealed opaque envelope bearing the name and address of the bidder, contractor's license number, identification of contract being bid and name of project. All bids must comply with the laws of the State of South Carolina.

THE OWNER RESERVES THE RIGHT TO REJECT ANY AND ALL BIDS, WAIVE INFORMALITIES AND AWARD THE CONTRACT IN THE BEST INTEREST OF THE OWNER.

End of Section

AIA DOCUMENTS

The following AIA forms are available for review at the office of FW Architects Inc.

AIA A701 Instructions To Bidders
AIA A101 Standard Form of Agreement Between Owner and Contractor
AIA A201 General Conditions of the Contract for Construction
AIA A310 Bid Bond
AIA A312 Performance and Payment Bond
AIA G701 Change Order
AIA G702 Application and Certificate of Payment
AIA G703 Continuation Sheet
AIA G704 Certificate of Substantial Completion
AIA G705 List of Subcontractors
AIA G706 Contractor's Affidavit of Payment of Debts and Claims
AIA G706A Contractor's Affidavit of Release of Liens
AIA G707 Consent of Surety to Final Payment
AIA G707A Consent of Surety to Reduction in / or Partial Release of Retainage
AIA G715 Certificate of Insurance
AIA G716 Request for Information (RFI)
AIA G810 Submittal Transmittal Letter

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BID FORM

**NEW WOOD FLOORING AND
TELESCOPING BLEACHERS
FOR
JOHNSONVILLE HIGH SCHOOL
GYMNASIUM
JOHNSONVILLE, SC**

EACH BID SHALL BE SUBMITTED ON FORMS WORDED AS FOLLOWS:

DATED: _____

Dear Sir:

The undersigned, having carefully examined General Conditions, Modifications to General Conditions, Extension of General Conditions, Specifications, the Drawings, as well as the site and conditions affecting the work: Propose to furnish all labor, materials, equipment, tools and services necessary for and incidental to the erection and completion of the Project 2303.

BASE BID - TOTAL (\$ _____)

Written word Bid Amount _____

ALTERNATES:

ADD:

DEDUCT:

ALTERNATE #1

**Painting Exposed Gymnasium Ceiling
As specified**

\$ _____

\$ _____

ALTERNATE #2

If included in Addendum

\$ _____

\$ _____

7. LISTING OF PROPOSED SUBCONTRACTORS

Bidder shall provide Subcontractors in the performance of the Subcontractor

Classification work listed:

Column A

Column B

SUBCONTRACTOR CLASSIFICATION	SUBCONTRACTOR'S PRIME CONTRACTOR'S NAME <i>(Must be completed by Bidder)</i>	
Gym Flooring Manufacturer		
Telescoping Bleacher Manf.		

TIME OF COMPLETION:

The time of completion is of extreme importance. Work shall begin within 10 days of signed contracts and Notice to Proceed is issued.

In computing the Contract completion time, calendar days have been added to account for the days of delay normally expected to be encountered due to weather conditions during the contract construction period. Normally expected weather conditions include rainfall amounts up to 0.50" in a twenty-four hour period based on the nearest station reporting and published by the National Oceanic and Atmospheric Administration. Only abnormal or unusual weather conditions in excess of this will be considered. Note: 5 days per month have been added as normal weather days.

Unless extension of time be granted as provided for in the A.I.A. General Conditions, work must be substantially complete within one hundred eighty calendar days 60 days from receipt of the Notice To Proceed. Substantially complete being defined as having the various facets of the project ready and available for full use and occupancy by the Owner.

Note: See Allowance Specification Section 01 21 00 for allowances to be included in bid.

LIQUIDATED DAMAGES:

The Contractor agrees, on the basis of the contract time for substantial completion of the entire project that from the contract price the Owner may retain the sum of Two Hundred Fifty Dollars (\$250.00) per calendar day for each day thereafter that any of the work remains incomplete.

SPECIAL NOTE: I, THE UNDERSIGNED CONTRACTOR, HAVE VISITED THE SITE AND REVIEWED THE AREA WHERE THE STRUCTURE SHALL BE LOCATED AND NOTED EXISTING CONDITIONS.

ACKNOWLEDGMENT OF ADDENDA:

NUMBER AND DATE

NUMBER AND DATE

NUMBER AND DATE

NUMBER AND DATE

NUMBER AND DATE

NUMBER AND DATE

Respectfully submitted,

By: _____

General Contractor's No.: _____

End of Section

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SECTION 00 72 13 - GENERAL CONDITIONS

PART ONE - GENERAL:

- 1-01 A.I.A. GENERAL CONDITIONS:** The General Conditions of the Contract for the Construction Building, A.I.A., Document No. A201, Latest Edition, Article 1 to 14 inclusive, except as may be inconsistent herewith, are hereby made part of the contract documents to same extent as if herein written out in full. They may be examined at the Architect's office. Where any Article of the A.I.A. General Conditions is modified, provisions of such Article not specifically modified shall remain in effect.

The Contract for this project shall be AIA Document A101, Latest Edition.

1-02 MODIFICATIONS TO STANDARD FORM GENERAL CONDITIONS:

- A. ARTICLE 1.3 Copies Furnished:** Shall be modified as follows: **Two (2) sets** of Drawings and Specifications will be furnished to the General Contractor as per agreement; additional digital sets will be furnished at cost of reproduction.
- B. ARTICLE 7.5 Performance Bond and Labor and Material Payment Bond:** Each Contractor shall be required to furnish a 5% bid bond. The successful contractor shall be required to furnish a Performance Bond in the sum of 100% of the Contract amount and a Labor and Materials Payment Bond in the same amount (100%) of the contract amount. Bonds shall be issued on the latest Documents issued and/or approved by the American Institute of Architects for this purpose and shall be written by a Surety Company approved by the Owner.
- C. ARTICLE 9 - Payment and Completion:** Shall be modified as follows: In accordance with Article 9, payment shall be made each month for 90% of the value as found by the Architect of materials suitably stored and insured or in place and labor performed during the preceding month.
- D. ARTICLE 11.1 Contractor's Liability Insurance:** Shall be modified as follows: The Contractor shall provide Insurance Certificates as follows:
- (1) Workmen's Compensation Insurance.
 - (2) Public Liability Insurance covering Bodily Injury and Property Damage.
 - (3) Vehicle Liability Insurance covering Bodily Injury and Property Damage.
 - (4) Owner's and Contractor's Protective Liability Insurance covering Bodily Injury and Property Damage.

The Limits as follows:

- (1) \$1,000,000.00 for injury to any person and \$1,000,000.00 total damage in any one accident.
- (2) \$250,000.00 for Property Damage on account of one accident and \$1,000,000.00 for all accidents during performance of the contract.
- (3) \$1,000,000.00 combined single limit.

E. ARTICLE 11.3 PROPERTY INSURANCE (Contractor's and Sub-contractors Insurance, i.e. Builder's Risk Insurance) (Fire and Extended Coverage).

- (1) The property insurance (Builder's Risk) shall be carried and maintained by the Owner as an extension of his existing policy.
- (2) The Owner shall obtain a Certificate of Compliance and shall transmit same through the Architect to be furnished to the General Contractor.
- (3) The Owner's insurance has a "Deductible" on fire, extended coverage, vandalism and malicious mischief. The amount of this "Deductible" and any other losses not specifically covered under the policy shall be borne by the General Contractor. This insurance does not cover any loss from theft or burglary, nor does it cover loss of any tools, equipment, scaffolding, staging, towers, forms, machinery, etc. owned or rented by mechanics or the contractor or subcontractor, which are not intended to become a part of the project; but does cover damage to the building or contents because of theft or burglary.
- (4) The General Contractor must submit any claims against this Policy in writing to the Owner, through the Architect, within twenty-one (21) days after occurrence in order that damage be assessed before job conditions are disturbed.

F. THE GENERAL CONTRACTOR AND SUBCONTRACTORS INSURANCE: In addition to that insurance stated above, the general and each subcontractor shall carry insurance on all stored materials for theft to the value requested on stored materials stored at the job site to the protection of the general contractor and the Owner. Any material stored off the job site shall be required to have total coverage for fire, theft, etc. Certificate of coverage shall be delivered to the Architect prior to materials being paid for.

End of Section

SECTION 00 73 00 – EXTENSIONS OF GENERAL CONDITIONS

PART ONE - GENERAL:

1-01 ARTICLE 15, APPROVAL OF MATERIALS: The specifications are written in three categories:

- A. Open Specification.
- B. Or approved equal specification.
- C. Base Bid Specifications.

Which are defined as follows:

- A. Open Specification: Such as chemical composition, or reference by number, symbol or title of a specified standard, such as a Federal specification or other similar standard. Where this open specification is used, any product meeting said specification may be used.
- B. Or approved equal specification: Product referenced by name, make or catalog number followed by the phrase "or approved equal." Where this or approved equal specification is used, it shall be interpreted to mean a standard to quality and shall not limit competition. **WRITTEN APPROVAL TEN (10) DAYS BEFORE BID SHALL BE SECURED FROM THE ARCHITECT BEFORE SUBSTITUTION IS MADE.**
- C. Base Bid Specification: One, two or three products referenced by name, make or catalog number. Where this base bid specification is used, the product or one of the products shall be used in preparing and submitting the Bid.

1-02 ARTICLE 16, TEMPORARY HEAT AND WIRING AND WATER:

- A. The Contractor shall provide heat, fuel and services as necessary to protect all work and materials against injury from dampness and cold until final acceptance of all work and material on the Contract unless the building or buildings are fully occupied by the Owner prior to such acceptance, in which case, the Owner shall assume all expenses of heating from the date of occupancy.
- B. If the permanent heating system is not ready for use, the Contractor shall submit to the Architect for approval, the method type of temporary heat that he (the Contractor) proposes to use.
- C. Water: The general contractor shall be responsible for providing construction water during the construction phase. The General Contractor may use existing owner's water but must compensate the owner for water used. The amount will be

determined by comparison of the previous two years water bills with the present one.

1-03 ARTICLE 16, TEMPORARY ELECTRICAL SERVICE: The Contractor shall be responsible for furnishing necessary electrical lighting and electrical power for all trades to accomplish their work throughout the construction phase. All temporary wiring and lighting shall be completely removed when permanent wiring is installed and the permanent lighting system shall be used to complete the construction. All wiring must meet the requirements of OSHA and the NEC and be metered separately.

1-04 ARTICLE 17, CLIMATIC CONDITIONS: When so ordered by the Architect, the Contractor shall suspend any work that may be subject to damage by climatic conditions.

1-05 ARTICLE 18, SIGNS:

- A. When directed by the architect, the General Contractor shall be required to erect and maintain, at a point as shall be designated, a construction sign. Said sign shall be approximately 4' x 8' made of wood with a neat border, with wording as directed and as per approved sketch (furnished by the Architect).
- B. No contractor shall display on or about the premises any sign, trademark or other advertising device except as may be necessary to identify such storage sheds or material delivery as he may require.

1-06 ARTICLE 19, TEMPORARY TELEPHONE: The contractor shall maintain cellular service on site at all times that work is underway for the duration of the contract.

1-07 ARTICLE 20, COMPUTER and PRINTER: The contractor shall maintain, without extra cost to the owner, a jobsite computer and printer for duration of the contract. Digital access must be maintained 24/7 for the transfer of data.

1-08 ARTICLE 21, TEMPORARY OFFICE: The contractor shall provide an 8' x 10' minimum space with sufficient heating, lighting, air conditioning, and telephone line. This space shall be maintained throughout construction phase, to be used by Architect, his field clerk and the General Contractor's superintendent.

1-09 ARTICLE 22, SANITARY ARRANGEMENTS: The General Contractor shall provide on the premise's suitable toilet facilities. These toilets shall be accessible to the workmen of all contractors. Construction personnel shall be instructed that existing facilities are not accessible.

1-10 ARTICLE 23, CONDITIONS AT SITE OF BUILDING:

- A. Bidders should visit the site and shall be responsible for having ascertained pertinent local conditions such as location, accessibility, and general character at the site of building, the character and extent of existing work within or adjacent to the site, and any other work being performed thereon at the time of the submission of his bid.
 - B. If, in the performance of the Contract, sub-surface or latent conditions at the site are found to be materially different from those indicated by the drawings and specifications, or unknown conditions of an unusual nature are disclosed differing materially from the condition usually inherent in work of the character shown and specified, the attention of the Architect shall be called immediately to such conditions before they are disturbed. Upon such notice, or upon his own observations of such conditions, the Architect shall promptly make such changes in the drawings and specifications as he finds necessary to conform to the different conditions, any increase or decrease in the cost of the work resulting from such changes shall be adjusted as provided under "Change of Work."
 - C. Safety Precautions at Site: A 6'-0" construction fence shall be placed around all storage areas, construction areas, and hazardous areas. It shall be the contractor's responsibility to maintain a **SAFE** construction atmosphere at all times and meet all OSHA requirements. All construction areas shall be completely fenced off and all exits shall be maintained and be kept safe at all times to insure the security of the new facility and the safety of the staff and residents.
- 1-11 ARTICLE 24, ADDENDA:** Any addenda issued prior to the time of bidding shall be covered in the bid and shall become a part thereof and shall be binding as to final executed documents.
- 1-12 ARTICLE 25, PREPARATION AND SUBMISSION OF BIDS:**
- A. The Bidder is required to Bid on all items called for in the Bid form. If an alternate price called for does not involve a change in price, the Bidder shall so indicate by writing the words "No Change" in the space provided.
 - B. Bids shall be submitted on the forms furnished, or copies thereof and shall be signed in ink. Erasures or other changes in a Bid must be explained or noted over the signature of the Bidder. Bids containing any conditions, omissions, unexplained erasures or alternates, or items not called for in the proposal or irregularities of any kind may be rejected by the Owner/Architect as being incomplete.
 - C. Each Bid must give the full business address of the Bidder and be signed by him with his usual signature. Bids by partnerships or joint ventures must furnish the full name of the parties and must be signed in the partnership by one of the members of the partnership or by an authorized representative, followed by the signature and designation of the person signing. Bids by Corporations must be signed with the

legal name of the Corporation, followed by the name of the State of Incorporation and by signature and designation of the President, Secretary or other person authorized to bind it in the matter. The name of each person signing shall also be typed or printed below the word "President," "Secretary," "Agent," or other designation, without disclosing his principal, may be held to the bid of the individual signing. When requested by the Owner, satisfactory evidence of the authority of the officer signing in behalf of the Corporation shall be furnished.

- D. Bids shall be enclosed in a sealed envelope which shall be marked and addressed as required by directions for mailing on the Bid Form.

1-13 ARTICLE 26, RECEIPT AND OPENING OF BIDS:

- A. Bids will be opened publicly at the time and place stated in the Invitation for Bids. The officers whose duty it is to open them will decide when the specified time has arrived and no Bid received thereafter will be considered. No responsibility will attach to any officer for the premature opening of a Bid not properly addressed and identified.
- B. Telegraphic Bids will not be considered, but modifications by telegraph of Bids already submitted will be considered if received prior to the hour set for opening.

1-14 ARTICLE 27, AWARD OF CONTRACT:

- A. The Contract will be awarded as soon as possible after 16 days have elapsed or within thirty (30) days to the lowest responsible Bidder, provided his Bid is reasonable and it is in the best interest of the Owner to accept it.
- B. The Owner reserves the right to waive any informality in Bids received and to reject any and all Bids when such waiver or rejection is in the interest of the Owner, and to award the job in the best interest of the Owner.
- C. Each Bidder shall be prepared, if so requested by the Owner, to present evidence of his experience, educational qualifications and financial ability to carry out the terms of the Contract.
- D. The Contract shall be issued on the latest AIA Document, Standard Form of Agreement Between Owner and Contractor.

- 1-15 AS BUILT DRAWINGS:** The General Contractor and all prime subcontractors shall, at the completion of the project and prior to final payment, furnish to the Architect a red-line set of prints with all changes to reflect as built drawings. The Architect shall then transpose these to the originals and furnish to the Owner a complete set of as built drawings.

- 1-16 RELEASE OF LIENS:** Contractor to execute and furnish this form to the Architect at completion of the project.
- 1-17 CONSENT OF SURETY TO FINAL PAYMENT:** (AIA G-707) Contractor to furnish this executed form to the Architect at the completion of the project.
- 1-18 BUILDING CODE REQUIREMENT:** The General Contractor shall stencil in red above the ceiling at all fire wall locations the following notation: "Fire and Smoke Barrier, Protect all Openings" with the appropriate hour rating notation and a solid red line between stencils. This shall be installed in locations as directed by the architect. Stencil shall be located every 15'-0" on center.
- 1-19 DAVIS-BACON ACT REQUIREMENTS:** This project is subject to the Davis-Bacon Act for wage requirements and federal government wage reporting.
- 1-20 CHANGE ORDERS:** No deviation from the construction documents shall be made without a written change order agreed upon and signed by the General Contractor, Architect and the owner. No monies will be paid for any change order until the documents have been processed and signed by all parties.

1-21 AIA FORMS:

AIA A701 Instructions To Bidders
AIA A101 Standard Form of Agreement Between Owner and Contractor
AIA A201 General Conditions of the Contract for Construction
AIA A310 Bid Bond
AIA A312 Performance and Payment Bond
AIA G701 Change Order
AIA G702 Application and Certificate of Payment
AIA G703 Continuation Sheet
AIA G704 Certificate of Substantial Completion
AIA G705 List of Subcontractors
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AIA G706A Contractor's Affidavit of Release of Liens
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AIA G715 Certificate of Insurance
AIA G716 Request for Information (RFI)
AIA G810 Submittal Transmittal Letter

End of Section

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DIVISION NO. 01 – GENERAL REQUIREMENTS OF THE CONTRACT

SECTION 01 21 00 - ALLOWANCES

PART ONE - GENERAL:

1-01 RELATED DOCUMENTS:

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1-02 SUMMARY:

- A. This Section includes administrative and procedural requirements governing allowances.
 - 1. Selected materials and equipment are specified in the Contract Documents by allowances. In some cases, these allowances include installation. Allowances have been established in lieu of additional requirements and to defer selection of actual materials and equipment to a later date when additional information is available for evaluation. If necessary, additional requirements will be issued by Change Order.
- B. Types of allowances include the following:
 - 1. Lump-sum allowances.
 - 2. Unit-cost allowances.
 - 3. Contingency allowances.
 - 4. Inspection and testing allowances.
- C. Schedule of Allowances:
 - 1. General Contingency to be used by owner as necessary: **\$ 30,000.00**
 - 2. Asbestos Removal Allowance: **\$35,000.00**
 - 3. Division 1 Section "Modification Procedures" specifies procedures for submitting and handling Change Orders.
 - 4. Division 1 Section "Quality Control Services" specifies procedures governing the use of allowances for inspection and testing.

1-03 SELECTION AND PURCHASE:

- A. At the earliest practical date after award of the Contract, advise the Architect of the date when the final selection and purchase of each product or system described by an allowance must be completed to avoid delaying the Work.

- B. At the Architect's request, obtain proposals for each allowance for use in making final selections. Include recommendations that are relevant to performing the Work.
- C. Purchase products and systems selected by the Architect from the designated supplier.

1-04 SUBMITTALS:

- A. Submit proposals for purchase of products or systems included in allowances, in the form specified for Change Orders.
- B. Submit invoices or delivery slips to show the actual quantities of materials delivered to the site for use in fulfillment of each allowance.

1-05 INSPECTION AND TESTING ALLOWANCES:

- A. Inspection and testing allowances include the cost of engaging the inspection or testing agencies, the actual inspections and tests, and reporting the results.
- B. The allowance does not include incidental labor required to assist the testing agency or costs for retesting upon failure of previous tests and inspections.
- C. Costs of services not required by the Contract Documents are not included in the allowance.
- D. At Project closeout, credit unused amounts remaining in the inspection and testing allowance to Owner by Change Order.

1-06 UNUSED MATERIALS:

- A. Return unused materials to the manufacturer or supplier for credit to the Owner, after installation has been completed and accepted.
 - 1. When requested by the Architect, prepare unused material for storage by Owner where it is not economically practical to return the material for credit. When directed by the Architect, deliver unused material to the Owner's storage space. Otherwise, disposal of unused material is the Contractor's responsibility.

PART TWO – PRODUCTS (Not Applicable)

PART THREE – EXECUTION:

3-01 EXAMINATION:

- A. Examine products covered by an allowance promptly upon delivery for damage or defects.

3-02 PREPARATION:

- A. Coordinate materials and their installation for each allowance with related materials and installations to ensure that each allowance item is completely integrated and interfaced with related work.

End of Section

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SECTION 01 23 00 – ALTERNATES

PART ONE - GENERAL:

- 1-01 REQUIREMENTS:** The requirements of all section of Division One apply to work included under this section.
- 1-02 WORK INCLUDED:** Furnish all labor and materials necessary to complete alternates indicated, specified or both.
- 1-03 ALTERNATE PRICE:** Each General Contractor shall quote in his bid the amount to be added or deducted to accomplish the Alternate listed below.

PART TWO - ALTERNATES:

- 2-01 ALTERNATE NUMBER ONE:** Contractor shall provide a cost for labor and material for all work associated with prepping and preparing the exposed roof deck, bar joists,. conduit and ductwork to receive minimum of 2 coats (or as required for total coverage) of white Dryfall Paint prior to installing new gym flooring and bleachers.
- 2-02 ALTERNATE NUMBER TWO:** If included in Addendum

END OF SECTION

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SECTION 01 25 13 - PRODUCT SUBSTITUTIONS

PART ONE - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and other Division-00 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section specifies administrative and procedural requirements for handling requests for substitutions made after award of the Contract.
 - 1. Multiple Prime Contracts: Provisions of this Section apply to the construction activities of each prime Contractor.
- B. The Contractor's Construction Schedule and the Schedule of Submittals are included under Section "Submittals."
- C. Standards: Refer to Section "Definitions and Standards" for applicability of industry standards to products specified.
- D. Procedural requirements governing the Contractor's selection of products and product options are included under Section "Materials and Equipment."

1.3 DEFINITIONS

- A. Definitions used in this Article are not intended to change or modify the meaning of other terms used in the Contract Documents.
- B. Substitutions: Requests for changes in products, materials, equipment, and methods of construction required by Contract Documents proposed by the Contractor after award of the Contract are considered requests for "substitutions." The following are not considered substitutions:
 - 1. Substitutions requested by Bidders during the bidding period, and accepted prior to award of Contract, are considered as included in the Contract Documents and are not subject to requirements specified in this Section for substitutions.
 - 2. Revisions to Contract Documents requested by the Owner or Architect.
 - 3. Specified options of products and construction methods included in Contract Documents.
 - 4. The Contractor's determination of and compliance with governing regulations and orders issued by governing authorities.

1.4 **SUBMITTALS**

- A. Substitution Request Submittal: Requests for substitution will be considered if received within 60 days after commencement of the Work. Requests received more than 60 days after commencement of the Work may be considered or rejected at the discretion of the Architect.
1. Submit 3 copies of each request for substitution for consideration. Submit requests in the form and in accordance with procedures required for Change Order proposals.
 2. Identify the product, or the fabrication or installation method to be replaced in each request. Include related Specification Section and Drawing numbers. Provide complete documentation showing compliance with the requirements for substitutions, and the following information, as appropriate:
 - a. Product Data, including Drawings and descriptions of products, fabrication and installation procedures.
 - b. Samples, where applicable or requested.
 - c. A detailed comparison of significant qualities of the proposed substitution with those of the Work specified. Significant qualities may include elements such as size, weight, durability, performance and visual effect.
 - d. Coordination information, including a list of changes or modifications needed to other parts of the Work and to construction performed by the Owner and separate Contractors, that will become necessary to accommodate the proposed substitution.
 - e. A statement indicating the substitution's effect on the Contractor's Construction Schedule compared to the schedule without approval of the substitution. Indicate the effect of the proposed substitution on overall Contract Time.
 - f. Cost information, including a proposal of the net change, if any in the Contract Sum.
 - g. Certification by the Contractor that the substitution proposed is equal-to or better in every significant respect to that required by the Contract Documents, and that it will perform adequately in the application indicated. Include the Contractor's waiver of rights to additional payment or time, that may subsequently become necessary because of the failure of the substitution to perform adequately.
 3. Architect's Action: Within one week of receipt of the request for substitution, the Architect will request additional information or documentation necessary for evaluation of the request. Within 2 weeks of receipt of the request, or one week of receipt of the additional information or documentation, which ever is later, the Architect will notify the Contractor of acceptance or rejection of the proposed substitution. If a decision on use of a proposed substitute cannot be made or obtained within the time allocated, use the product specified by name. Acceptance will be in the form of a Change Order.

PART TWO - PRODUCTS

2.1 SUBSTITUTIONS

- A. Conditions: The Contractor's substitution request will be received and considered by the Architect when one or more of the following conditions are satisfied, as determined by the Architect; otherwise requests will be returned without action except to record noncompliance with these requirements.
1. Extensive revisions to Contract Documents are not required.
 2. Proposed changes are in keeping with the general intent of Contract Documents.
 3. The request is timely, fully documented and properly submitted.
 4. The request is directly related to an "or equal" clause or similar language in the Contract Documents.
 5. The specified product or method of construction cannot be provided within the Contract Time. The request will not be considered if the product or method cannot be provided as a result of failure to pursue the Work promptly or coordinate activities properly.
 6. The specified product or method of construction cannot receive necessary approval by a governing authority, and the requested substitution can be approved.
 7. A substantial advantage is offered the Owner, in terms of cost, time, energy conservation or other considerations of merit, after deducting offsetting responsibilities the Owner may be required to bear. Additional responsibilities for the Owner may include additional compensation to the Architect for redesign and evaluation services, increased cost of other construction by the Owner or separate Contractors, and similar considerations.
 8. The specified product or method of construction cannot be provided in a manner that is compatible with other materials, and where the Contractor certifies that the substitution will overcome the incompatibility.
 9. The specified product or method of construction cannot be coordinated with other materials, and where the Contractor certifies that the proposed substitution can be coordinated.
 10. The specified product or method of construction cannot provide a warranty required by the Contract Documents and where the Contractor certifies that the proposed substitution provide the required warranty.
 11. Where a proposed substitution involves more than one prime Contractor, each Contractor shall cooperate with the other Contractors involved to coordinate the Work, provide uniformity and consistency, and to assure compatibility of products.
- B. The Contractor's submittal and Architect's acceptance of Shop Drawings, Product Data or Samples that relate to construction activities not complying with the Contract Documents does not constitute an acceptable or valid request for substitution, nor does it constitute approval.

PART THREE - EXECUTION (Not Applicable)

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SECTION 01 26 00 – CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1-01 DESCRIPTION

- A. NO EXTRA WORK SHALL BE PERFORMED WITHOUT FIRST RECEIVING WRITTEN APPROVAL FROM THE ARCHITECT.
- B. Work included: Make such changes in the Work, in the Contract Sum, in the Contract Time of Completion, to any combination thereof, as are described in written Construction Change Directives or written Change Orders signed by the Owner and the Architect and issued after execution of the Contract, in accordance with provisions of this Section.
- C. Related Work:
 - 1. Documents affecting work of this Section include, but are not necessarily limited to, AIA Document A201 and Spec Sections 00 72 13-General Conditions and 00 73 00-Extension of General Conditions, found in Division 00 of these Specifications.
 - 2. Changes in the Work are described further in Article 7 of the General Conditions.

1-02 QUALITY ASSURANCE

- A. Include within the Contractor's quality assurance program such measures as are needed to assure familiarity of the Contractor's staff and employees with these procedures for processing Change Order data.

1-03 SUBMITTALS

- A. Make submittals directly to the Architect.
- B. Submit the number of copies called for under the various items listed in this Section along with appropriate back-up materials.

1-04 PROCESSING CHANGES INITIATED BY THE OWNER

- A. Should the Owner contemplate making a change in the Work or a change in the Contract Time of Completion, the Architect will issue a Proposed Change Order request or a Construction Change Directive to the Contractor.

1. Proposed Change Order requests will describe the contemplated change and will request that the Contractor provide a detailed price and estimate of time that it seeks in the event the change is authorized by the Owner.
 2. Construction Change Directives will be dated and will be numbered in sequence.
 3. The Construction Change Directives will describe the contemplated change, and will carry one of the following instructions to the Contractor:
 - a. Make the described change in the Work at no change in the Contract Sum and no change in the Contract Time of Completion in accordance with Paragraph 7.3 of the General Conditions.
 - b. Make the described change in the Work, and provide for a credit or cost to be determined in accordance with Paragraph 7.3.7 of the General Conditions.
- B. If the Contractor has been directed by the Architect to make the described change in the Work at no change in the Contract Sum and no change in the Contract Time of Completion, but the Contractor wishes to make a claim for one or both of such changes, the Contractor shall proceed with the change and shall notify the Architect as provided for under Article 15.1 of the General Conditions.
- C. If the Contractor has been directed by the Architect to make the described change subject to later determination of cost or credit in accordance with Paragraph 7.3 of the General Conditions, the Contractor shall:
1. Take such measures as needed to make the change;
 2. Consult with the Architect and reach agreement on the most appropriate method for determining credit or cost for the change.
- D. If the Contractor has been directed by the Architect to promptly advise him as to credit or cost proposed for the described change, the Contractor shall:
1. Analyze the described change and its impact on costs and time;
 2. Secure the required information and forward it to the Architect for review;
 3. Meet with the Architect as required to explain costs and, when appropriate, determine other acceptable ways to achieve the desired objective;
 4. Alert pertinent personnel and subcontractors as to the impending change and, to the maximum extent possible, avoid such work as would increase the Owner's cost for making the change, advising the Architect in writing when avoidance no longer is practicable.

1-05 PROCESSING CHANGES INITIATED BY THE CONTRACTOR

If the Contractor recommends a change in the Work, he shall submit a Proposed Change Order with detailed cost and time information as detailed below.

1-06 PROCESSING CHANGE ORDERS

CONTRACTOR'S ACTIONS

- A. Make written reply to the Architect in response to each Proposed Change Order request or Construction Change Directive.
 - 1. State proposed change in the Contract Sum, if any.
 - 2. State proposed change in the Contract Time of completion, if any.
 - 3. Clearly describe other changes in the Work required by the proposed change or desirable therewith, if any.
 - 4. Include full backup data such as, subcontractor's letter of proposal or similar information.
 - 5. Submit this response in a single copy.

ARCHITECT'S ACTIONS

- A. When cost of credit for the change has been agreed upon by the Owner and the Contractor, or the Owner has directed that cost or credit be determined in accordance with provisions of Paragraph 7.3 of the General Conditions, the Architect will issue a Change Order or Constructive Change Directive to the Contractor.
- B. Change Orders will be dated and will be numbered in sequence.
- C. The Change Order will describe the change or changes, will refer to the Construction Change Directive(s) involved, and will be signed by the Owner, and the Architect.
- D. The Architect will issue three copies of each Change Order to the Contractor.
 - 1. The Contractor promptly shall sign all three copies and return all three copies to the Architect for the Architect's signature.
 - 2. The Architect will sign all three copies and return three copies to the Owner for the Owner's signature.
 - 3. The Owner will sign all three copies, retain one copy for his file and return the remaining two copies to the Architect who will then forward fully executed copies to the Contractor.
- E. Should the Contractor disagree with the stipulated change in Contract Sum, or change in Contract Time of Completion, or both:
 - 1. The Contractor promptly shall return three copies of the Change Order, unsigned by him, to the Architect with a letter signed by the Contractor's disagreement.
 - 2. The Contractor's disagreement with the Change Order shall not in any way relieve the Contractor of his responsibility to proceed with the change as ordered under pertinent provisions of the Contract Documents.

End of Section

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SECTION 01 29 00 – PAYMENT PROCEDURES

PART ONE - GENERAL

1-01 DESCRIPTION

- A. Work included: Comply with procedures described in this Section when applying for progress payment and final payment under the Contract.
- B. Related Work:
 - 1. Documents affecting work of this Section include, but are not necessarily limited to, AIA Document A201 and Spec Sections 00 72 13-General Conditions and 00 73 00-Extension of General Conditions, found in Division 00 of these Specifications.
 - 2. The Contract Sum and the schedule for payments are described in the Form of Agreement.
 - 3. Payments upon Substantial Completion and Completion of the Work are described in the AIA General Conditions, as amended and in Division 01 of these Specifications, including - General Requirements-01 77 00 Execution and Closeout Procedures.
 - 4. The Architect's approval of applications for progress payment and final payment is contingent upon the Architect's approval of status of Project Record Documents pursuant to the requirements of Section 017839 and the AIA General Conditions, as amended.
 - 5. The Architect's approval of applications for progress payment is contingent upon receipt of Project Schedule updates to be submitted along with each pay application.

1-02 QUALITY ASSURANCE

- A. During progress of the Work, modify the schedule of values as approved by the Architect to reflect changes in the Contract Sum due to Change Orders or other modifications of the Contract.
- B. Base requests for payment on the approved schedule of values.
- C. Payment Application Times: Each progress payment date is as indicated in the Agreement. The period of construction Work covered by each Application or Payment is the period indicated in the Agreement.

- D. Application Preparation: complete every entry on the form, including notarization and execution by person authorized to sign legal documents on behalf of the Contractor. Incomplete applications will be returned without action.

1-03 SUBMITTALS

- A. Formal Submittal: Unless otherwise directed by the Architect.
1. Make formal submittal of request for payment by filling in all appropriate information, by typewriter or neat lettering in ink, on AIA Document G702, "Application and Certificate for Payment", plus Continuation Sheets, AIA Document G703, using data from the Schedule of Values and the accompanying cost loaded schedule, if applicable. Submit other supporting documentation required by the Architect as detailed by the AIA Documents including, but not limited to, lien waivers, Consent of Surety, etc.
 2. Sign and notarize the Application and Certificate for Payment.
 3. Submit the original of the Application and Certificate for Payment, plus two (2) identical copies of the entire Application including all continuation sheet or sheets, to the Architect. All copies shall bear original signatures and original notarizations.
 4. The Architect will review the formal submittal and, upon approval and agreement, will:
 - a. Retain one copy for the Architect's files and transmit two copies to the Owner.
 5. Request for Payment against any change order will not be honored until the change order is signed by all appropriate parties.
- B. Application for Payment at Substantial Completion: Following issuance of the Certificate of Substantial Completion, submit an Application for Payment; this application shall reflect any Certificates of Partial Substantial Completion issued previously for Owner occupancy or designated portions of the Work.

Administrative actions and submittal that shall proceed or coincide with this application include:

1. Occupancy permits and similar approvals.
2. Warranties (guarantees) and maintenance agreements.
3. Test/adjust/balance records.
4. Maintenance instructions.
5. Start-up performance reports.
6. Change-over information related to Owner's occupancy, use operation and maintenance.
7. Final cleaning.
8. Application for reduction of retainage and consent of surety.
9. Advice on shifting insurance coverages.

10. Punch list of incomplete Work, recognized as exception to Architect's Certificate of Substantial Completion (area by area).
 11. Change of door locks to Owner's access keys.
- C. Final Payment Application: In conjunction with the requirements of administrative actions and submittals which must precede or coincide with submittal of the final payment Application for Payment include the following:
1. Completion of Project closeout requirements.
 2. Completion of items specified for completion after Substantial Completion.
 3. A final release which indicates no further claims will be submitted against this contract.
 4. Assurance that Work is complete.
 5. Transmittal of required Project Construction Records to Owner including all record documents.
 6. Proof that all taxes, fees and similar obligations have been paid.
 7. All required lien releases/waiver of claims from subcontractors, suppliers and other vendors.
 8. Consent of Surety.

END OF SECTION

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SECTION 01 32 00 - CONSTRUCTION PROGRESS DOCUMENTATION

PART ONE - GENERAL

1-01 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General Conditions and other Division 1 Specification Sections, apply to this Section.

1-02 SUMMARY

- A. This Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:
 - 1. Schedule. (See Section 01 32 16)
 - 2. Submittals Schedule.
 - 3. Daily construction reports.
 - 4. Material location reports.
 - 5. Field condition reports.
 - 6. Special reports.
- B. Specific requirements of each contract are also indicated in individual Specification Sections, All Bid Documents and on Drawings.
- C. Daily Construction Reports: Submitted Daily to architect & Owner, in a format as defined by Owner.
- D. Material Location Reports: Submit as required by architect & Owner.
- E. Field Condition Reports: Submitted to Architect & Owner, in a format as defined by architect & Owner, (including electronic formats such as Primavera Expedition).
- F. Special Reports: Submitted per occurrence to Architect and Owner's Representative in a format as defined by architect & Owner (including electronic formats such as Primavera Expedition).

1-03 QUALITY ASSURANCE

- 1. Photographer Qualifications: Digital photos of high quality taken by the General Contractor are acceptable.

1-04 COORDINATION

- A. Auxiliary Services: Cooperate with other trades, architect & Owner, and provide auxiliary services requested, including access to Project site and use of temporary facilities including temporary lighting.

PART 2 - PRODUCTS

2-01 SUBMITTALS SCHEDULE

- A. Preparation: Submit a schedule of submittals, arranged in chronological order by dates required by construction schedule. Include time required for review, resubmittal, ordering, manufacturing, fabrication, and delivery when establishing dates.
 - 1. Coordinate Submittals Schedule with list of subcontracts, the Schedule of Values, and Contractor's Construction Schedule.
 - 2. Initial Submittal: Submit concurrently with preliminary network diagram. Include submittals required during the first 60 days of construction. List those required to maintain orderly progress of the Work and those required early because of long lead-time for manufacture or fabrication.
 - a. Show submittals on the Preliminary Construction Schedule.
 - b. Submittals must be logged and maintained in a format as defined by architect & Owner (including electronic formats such as Primavera Expedition).

2-02 CONTRACTOR'S CONSTRUCTION SCHEDULE, GENERAL

(For specific information – refer to Specification Section 01 32 16)

- 1. Phasing: Arrange list of activities on schedule by phase.
- 2. Work under More Than One Contract: Include a separate activity for each contract.
- 3. Work by The Owner: Include a separate activity for each portion of the Work performed by The Owner.
- 4. Products Ordered in Advance: Include a separate activity for each product. Include delivery date. Delivery dates indicated stipulate the earliest possible delivery date.
- 5. The Owner-Furnished Products: Include a separate activity for each product. Include delivery date. Delivery dates indicated stipulate the earliest possible delivery date.
- 6. Work Restrictions: Show the effect of the following items on the schedule:
 - a. Coordination with existing construction.
 - b. Limitations of continued occupancies.
 - c. Uninterruptible services.
 - d. Partial occupancy before Substantial Completion. (Show staff occupying the building to set up classes & stocking at least 30 days prior to the contract substantial completion date).

- e. Use of premises restrictions.
 - f. Provisions for future construction.
 - g. Seasonal variations.
 - h. Environmental control.
7. Work Stages: Indicate important stages of construction for each major portion of the Work, including, but not limited to, the following:
- a. Subcontract awards.
 - b. Submittals.
 - c. Purchases.
 - d. Mockups.
 - e. Fabrication.
 - f. Sample testing.
 - g. Deliveries.
 - h. Installation.
 - i. Tests and inspections.
 - j. Adjusting.
 - k. Curing.
 - l. Startup and placement into final use and operation.
8. Area Separations: Identify each major area of construction for each major portion of the Work. Indicate where each construction activity within a major area must be sequenced or integrated with other construction activities to provide for the following:
- a. Structural completion.
 - b. Permanent space enclosure.
 - c. Completion of mechanical installation.
 - d. Completion of electrical installation.
 - e. Substantial Completion.
9. Responsibilities: Identify each activity according to the responsibility for that activity. Responsibilities categorization of activities shall include
- a. The Owner
 - b. Architect
 - c. City or County Agency having jurisdiction
 - d. General Contractor
 - e. Mechanical Contractor
 - f. Plumbing Contractor
 - g. HVAC Contractor
 - h. Electrical Contractor
 - i. Technology Contractor
 - j. Others having prime contracts

The purpose of this responsibility is to sort the schedule by entities having prime agreements with The Owner, Architect, agencies having jurisdiction. Establish secondary responsibilities in a separate activity definition for the purposes of sorting by subcontractors for the contractor's convenience.

2-03 REPORTS

- A. Daily Construction Reports: Prepare a daily construction report recording the following information concerning events at Project site:

1. List of subcontractors at Project site.
 2. List of separate contractors at Project site.
 3. Approximate count of personnel at Project site.
 4. High and low temperatures and general weather conditions.
 5. Accidents.
 6. Meetings and significant decisions.
 7. Unusual events (refer to special reports).
 8. Stoppages, delays, shortages, and losses.
 9. Meter readings and similar recordings.
 10. Emergency procedures.
 11. Orders and requests of authorities having jurisdiction.
 12. Change Orders received and implemented.
 13. Construction Change Directives received.
 14. Services connected and disconnected.
 15. Equipment or system tests and startups.
 16. Partial Completions and occupancies.
 17. Substantial Completions authorized.
- B. Material Location Reports: At intervals as required by the architect & Owner, prepare a comprehensive list of materials delivered to and stored at Project site. List shall be cumulative, showing materials previously reported plus items recently delivered. Include with list a statement of progress on and delivery dates for materials or items of equipment fabricated or stored away from Project site.
- C. Field Condition Reports: Immediately on discovery of a difference between field conditions and the Contract Documents, prepare a detailed report. Submit with a request for information on Primavera Expedition. Include a detailed description of the differing conditions, together with recommendations for changing the Contract Documents.

2-04 SPECIAL REPORTS

- A. General: Submit special reports directly to architect & Owner, within one day of an occurrence. Distribute copies of report to parties affected by the occurrence by way of Primavera Expedition.
- B. Reporting Unusual Events: When an event of an unusual and significant nature occurs at Project site, whether or not related directly to the Work, prepare and submit a special report. List chain of events, persons participating, and response by Contractor's personnel, evaluation of results or effects, and similar pertinent information. Advise architect & Owner in advance when these events are known or predictable.

End of Section

SECTION 01 33 00 - SUBMITTALS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and other Division-1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section specifies administrative and procedural requirements for submittals required for performance of the Work, including;
1. Contractor's construction schedule.
 2. Submittal schedule.
 3. Daily construction reports.
 4. Shop Drawings.
 5. Product Data.
 6. Samples.
- B. Administrative Submittals: Refer to other Division-1 Sections and other Contract Documents for requirements for administrative submittals. Such submittals include, but are not limited to:
1. Permits.
 2. Applications for payment.
 3. Performance and payment bonds.
 4. Insurance certificates.
 5. List of Subcontractors.
- C. The Schedule of Values submittal is included in Section "Applications for Payment."
- D. Inspection and test reports are included in Section "Quality Control Services."
- E. Submittal of Project photographs is included under Section "Construction Photographs."

1.3 SUBMITTAL PROCEDURES

- A. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Transmit each submittal sufficiently in advance of performance of related construction activities to avoid delay.
1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals and related activities that require sequential activity.
 2. Coordinate transmittal of different types of submittals for related elements of the Work so processing will not be delayed by the need to review submittals

concurrently for coordination.

- a. The Architect reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
3. Processing: Allow sufficient review time so that installation will not be delayed as a result of the time required to process submittals, including time for resubmittals.
 - a. Allow two weeks for initial review. Allow additional time if processing must be delayed to permit coordination with subsequent submittals. The Architect will promptly advise the Contractor when a submittal being processed must be delayed for coordination.
 - b. If an intermediate submittal is necessary, process the same as the initial submittal.
 - c. Allow two weeks for reprocessing each submittal.
 - d. No extension of Contract Time will be authorized because of failure to transmit submittals to the Architect sufficiently in advance of the Work to permit processing.

B. Submittal Preparation: Place a permanent label or title block on each submittal for identification. Indicate the name of the entity that prepared each submittal on the label or title block.

1. Provide a space approximately 4" x 5" on the label or beside the title block on Shop Drawings to record the Contractor's review and approval markings and the action taken.
2. Include the following information on the label for processing and recording action taken.
 - a. Project name.
 - b. Date.
 - c. Name and address of Architect.
 - d. Name and address of Contractor.
 - e. Name and address of subcontractor.
 - f. Name and address of supplier.
 - g. Name of manufacturer.
 - h. Number and title of appropriate Specification Section.
 - i. Drawing number and detail references, as appropriate.

C. Submittal Transmittal: Package each submittal appropriately for transmittal and handling. Transmit each submittal from Contractor to Architect using a transmittal form. Submittals received from sources other than the Contractor will be returned without action.

1. On the transmittal Record relevant information and requests for data. On the form, or separate sheet, record deviations from Contract Document requirements, including minor variations and limitations. Include Contractor's certification that information complies with Contract Document requirements.
2. Transmittal Form: Use AIA Document G 810.
3. Transmittal Form: Use the sample form at the end of this Section for transmittal of submittals.

1.4 CONTRACTOR'S CONSTRUCTION SCHEDULE

- A. Bar-Chart Schedule: Prepare a fully developed, horizontal bar- chart type Contractor's construction schedule. Submit within 30 days of the date established for "Commencement of the Work".
1. Provide a separate time bar for each significant construction activity. Provide a continuous vertical line to identify the first working day of each week. Use the same breakdown of units of the Work as indicated in the "Schedule of Values".
 2. Within each time bar indicate estimated completion percentage in 10 percent increments. As Work progresses, place a contrasting mark in each bar to indicate Actual Completion.
 3. Prepare the schedule on a sheet, or series of sheets, of stable transparency, or other reproducible media, of sufficient width to show data for the entire construction period.
 4. Secure time commitments for performing critical elements of the Work from parties involved. Coordinate each element on the schedule with other construction activities; include minor elements involved in the sequence of the Work. Show each activity in proper sequence. Indicate graphically sequences necessary for completion of related portions of the Work.
 5. Coordinate the Contractor's construction schedule with the schedule of values, list of subcontracts, submittal schedule, progress reports, payment requests and other schedules.
 6. Indicate completion in advance of the date established for Substantial Completion. Indicate Substantial Completion on the schedule to allow time for the Architect's procedures necessary for certification of Substantial Completion.
- B. Phasing: Provide notations on the schedule to show how the sequence of the Work is affected by requirements for phased completion to permit Work by separate Contractors and partial occupancy by the Owner prior to Substantial Completion.
- C. Work Stages: Indicate important stages of construction for each major portion of the Work, including testing and installation.
- D. Area Separations: Provide a separate time bar to identify each major construction area for each major portion of the Work. Indicate where each element in an area must be sequenced or integrated with other activities.
- E. Cost Correlation: At the head of the schedule, provide a two item cost correlation line, indicating "precalculated" and "actual" costs. On the line show dollar-volume of Work performed as of the dates used for preparation of payment requests.
1. Refer to Section "Applications for Payment" for cost reporting and payment procedures.
- F. Distribution: Following response to the initial submittal, print and distribute copies to the Architect, Owner, subcontractors, and other parties required to comply with

scheduled dates. Post copies in the Project meeting room and temporary field office.

1. When revisions are made, distribute to the same parties and post in the same locations. Delete parties from distribution when they have completed their assigned portion of the Work and are no longer involved in construction activities.

G. Schedule Updating: Revise the schedule after each meeting or activity, where revisions have been recognized or made. Issue the updated schedule concurrently with report of each meeting.

1.5 SUBMITTAL SCHEDULE

A. After development and acceptance of the Contractor's construction schedule, prepare a complete schedule of submittals. Submit the schedule within 10 days of the date required for establishment of the Contractor's construction schedule.

1. Coordinate submittal schedule with the list of subcontracts, schedule of values and the list of products as well as the Contractor's construction schedule.
2. Prepare the schedule in chronological order; include submittals required during the first 90 days of construction. Provide the following information:
 - a. Scheduled date for the first submittal.
 - b. Related Section number.
 - c. Submittal category.
 - d. Name of subcontractor.
 - e. Description of the part of the Work covered.
 - f. Scheduled date for resubmittal
 - g. Scheduled date the Architect's final release or approval.

B. Distribution: Following response to initial submittal, print and distribute copies to the Architect, Owner, subcontractors, and other parties required to comply with submittal dates indicated. Post copies in the Project meeting room and field office.

1. When revisions are made, distribute to the same parties and post in the same locations. Delete parties from distribution when they have completed their assigned portion of the Work and are no longer involved in construction activities.

C. Schedule Updating: Revise the schedule after each meeting or activity, where revisions have been recognized or made. Issue the updated schedule concurrently with report of each meeting.

1.6 DAILY CONSTRUCTION REPORTS

A. Prepare a daily construction report, recording the following information concerning events at the site; and submit duplicate copies to the Architect at weekly intervals:

1. List of subcontractors at the site.
2. Approximate count of personnel at the site.
3. High and low temperatures, general weather conditions.

4. Accidents and unusual events.
5. Meetings and significant decisions.
6. Stoppages, delays, shortages, losses.
7. Meter readings and similar recordings.
8. Emergency procedures.
9. Orders and requests of governing authorities.
10. Change Orders received, implemented.
11. Services connected, disconnected.
12. Equipment or system tests and start-ups.
13. Partial Completions, occupancies.
14. Substantial Completions authorized.

1.7 SHOP DRAWINGS

- A. Submit newly prepared information, drawn to accurate scale. Highlight, encircle, or otherwise indicate deviations from the Contract Documents. Do not reproduce Contract Documents or copy standard information as the basis of Shop Drawings. Standard information prepared without specific reference to the Project is not considered Shop Drawings.
- B. Shop Drawings include fabrication and installation drawings, setting diagrams, schedules, patterns, templates and similar drawings. Include the following information:
 1. Dimensions.
 2. Identification of products and materials included.
 3. Compliance with specified standards.
 4. Notation of coordination requirements.
 5. Notation of dimensions established by field measurement.
 6. Sheet Size: Except for templates, patterns and similar full- size Drawings, submit Shop Drawings on sheets at least 8-1/2" x 11" but no larger than 36" x 48".
 7. Initial Submittal: Submit one correctable translucent reproducible print and one blue- or black-line print for the Architect's review; the reproducible print will be returned.
 8. Initial Submittal: Submit 2 blue- or black-line prints for the Architect's review; one will be returned.
 9. Final Submittal: Submit 3 blue- or black-line prints; submit 5 prints where required for maintenance manuals. 2 prints will be retained; the remainder will be returned.
 10. Final Submittal: Submit 3 blue- or black-line prints and 2 additional prints where required for maintenance manuals, plus the number of prints needed by the Architect for distribution. 2 prints will be retained; the remainder returned.
 - a. One of the prints returned shall be marked-up and maintained as a "Record Document".
 11. Do not use Shop Drawings without an appropriate final stamp indicating action taken in connection with construction.

- C. Coordination drawings are a special type of Shop Drawing that show the relationship and integration of different construction elements that require careful coordination during fabrication or installation to fit in the space provided or function as intended.
1. Preparation of coordination Drawings is specified in section "Project Coordination" and may include components previously shown in detail on Shop Drawings or Product Data.
 2. Submit coordination Drawings for integration of different construction elements. Show sequences and relationships of separate components to avoid conflicts in use of space.

1.8 PRODUCT DATA

- A. Collect Product Data into a single submittal for each element of construction or system. Product Data includes printed information such as manufacturer's installation instructions, catalog cuts, standard color charts, roughing-in diagrams and templates, standard wiring diagrams and performance curves. Where Product Data must be specially prepared because standard printed data is not suitable for use, submit as "Shop Drawings."
1. Mark each copy to show applicable choices and options. Where printed Product Data includes information on several products, some of which are not required, mark copies to indicate the applicable information. Include the following information:
 - a. Manufacturer's printed recommendations.
 - b. Compliance with recognized trade association standards.
 - c. Compliance with recognized testing agency standards.
 - d. Application of testing agency labels and seals.
 - e. Notation of dimensions verified by field measurement.
 - f. Notation of coordination requirements.
 2. Do not submit Product Data until compliance with requirements of the Contract Documents has been confirmed.
 3. Preliminary Submittal: Submit a preliminary single-copy of Product Data where selection of options is required.
 4. Submittals: Submit 2 copies of each required submittal; submit 4 copies where required for maintenance manuals. The Architect will retain one, and will return the other marked with action taken and corrections or modifications required.
 - a. Unless noncompliance with Contract Document provisions is observed, the submittal may serve as the final submittal.
 5. Distribution: Furnish copies of final submittal to installers, subcontractors, suppliers, manufacturers, fabricators, and others required for performance of construction activities. Show distribution on transmittal forms.
 - a. Do not proceed with installation until an applicable copy of Product Data applicable is in the installer's possession.
 - b. Do not permit use of unmarked copies of Product Data in connection with construction.

1.9 **SAMPLES**

- A. Submit full-size, fully fabricated Samples cured and finished as specified and physically identical with the material or product proposed. Samples include partial sections of manufactured or fabricated components, cuts or containers of materials, color range sets, and swatches showing color, texture and pattern.
1. Mount, display, or package Samples in the manner specified to facilitate review of qualities indicated. Prepare Samples to match the Architect's Sample. Include the following:
 - a. Generic description of the Sample.
 - b. Sample source.
 - c. Product name or name of manufacturer.
 - d. Compliance with recognized standards.
 - e. Availability and delivery time.
 2. Submit Samples for review of kind, color, pattern, and texture, for a final check of these characteristics with other elements, and for a comparison of these characteristics between the final submittal and the actual component as delivered and installed.
 - a. Where variation in color, pattern, texture or other characteristics are inherent in the material or product represented, submit multiple units (not less than 3), that show approximate limits of the variations.
 - b. Refer to other Specification Sections for requirements for Samples that illustrate workmanship, fabrication techniques, details of assembly, connections, operation and similar construction characteristics.
 - c. Refer to other Sections for Samples to be returned to the Contractor for incorporation in the Work. Such Samples must be undamaged at time of use. On the transmittal, indicate special requests regarding disposition of Sample submittals.
 3. Preliminary submittals: Where Samples are for selection of color, pattern, texture or similar characteristics from a range of standard choices, submit a full set of choices for the material or product.
 - a. Preliminary submittals will be reviewed and returned with the Architect's mark indicating selection and other action.
 4. Submittals: Except for Samples illustrating assembly details, workmanship, fabrication techniques, connections, operation and similar characteristics, submit 3 sets; one will be returned marked with the action taken.
 5. Maintain sets of Samples, as returned, at the Project site, for quality comparisons throughout the course of construction.
 - a. Unless noncompliance with Contract Document provisions is observed, the submittal may serve as the final submittal.
 - b. Sample sets may be used to obtain final acceptance of the construction associated with each set.
- B. Distribution of Samples: Prepare and distribute additional sets to subcontractors, manufacturers, fabricators, suppliers, installers, and others as required for performance of the Work. Show distribution on transmittal forms.

1. Field Samples specified in individual Sections are special types of Samples. Field Samples are full-size examples erected on site to illustrate finishes, coatings, or finish materials and to establish the standard by which the Work will be judged.
 - a. Comply with submittal requirements to the fullest extent possible. Process transmittal forms to provide a record of activity.

1.10 ARCHITECT'S ACTION

- A. Except for submittals for record, information or similar purposes, where action and return is required or requested, the Architect will review each submittal, mark to indicate action taken, and return promptly.
 1. Compliance with specified characteristics is the Contractor's responsibility.
- B. Action Stamp: The Architect will stamp each submittal with a uniform, self-explanatory action stamp. The stamp will be appropriately marked, as follows, to indicate the action taken:
 1. Final Unrestricted Release: Where submittals are marked "Approved," that part of the Work covered by the submittal may proceed provided it complies with requirements of the Contract Documents; final acceptance will depend upon that compliance.
 2. Final-But-Restricted Release: When submittals are marked "Approved as Noted," that part of the Work covered by the submittal may proceed provided it complies with notations or corrections on the submittal and requirements of the Contract Documents; final acceptance will depend on that compliance.
 3. Returned for Resubmittal: When submittal is marked "Not Approved, Revise and Resubmit," do not proceed with that part of the Work covered by the submittal, including purchasing, fabrication, delivery, or other activity. Revise or prepare a new submittal in accordance with the notations; resubmit without delay. Repeat if necessary to obtain a different action mark.
 - a. Do not permit submittals marked "Not Approved, Revise and Resubmit" to be used at the Project site, or elsewhere where Work is in progress.
 4. Other Action: Where a submittal is primarily for information or record purposes, special processing or other activity, the submittal will be returned, marked "Action Not Required".

PART 2 - PRODUCTS (Not Applicable).

PART 3 - EXECUTION (Not Applicable).

END OF SECTION

SECTION 01 41 00 – REGULATORY REQUIREMENTS

- A. The following requirements of Regulatory Agencies having an interest in this project are hereby made a part of this Contract.
- B. The construction of the project, including the letting of contracts in connection therewith, shall conform to the applicable requirements of State, territorial, and local laws and ordinances to the extent that such requirements do not conflict with Federal laws and this subchapter.
- C. South Carolina Sales Tax: All applicable South Carolina sales tax shall be to the account of the Contractor.
- D. Use of chemicals: All chemicals used during the project construction or furnished for project operation, whether herbicide, pesticide, disinfectant, polymer, reactant or of other classification, must show approval of EPA or USDA. Use of all such chemicals and disposal of residues shall be in strict conformance with instructions.
- E. Safety and Health Regulations: The Contractor shall comply with the Department of Labor and Safety and Health Regulations for construction promulgated under the Occupational Safety and Health Act of 1970 (PL-91-596) and under Section 107 of the Contract Work Hours and Safety Standards Act (PL91-54).
- F. Inspection by Agencies: The representatives of the South Carolina Department of Health and Environmental Control, Florence County and South Carolina Department of Highways and Public Transportation, and, where applicable, municipalities in which a project is located, shall have access to the work wherever it is, in preparation or in progress, and the Contractor shall provide proper facilities for such access and inspection.
- G. Withholding for Non-Residents shall comply with the following:
 - 1. Attention of non-resident contractors is invited to Part Two, Act No. 855, Acts of the General Assembly of South Carolina 1958.
 - 2. If a non-resident contractor is the successful bidder on this project, he shall be required to post surety bond, or deposit cash or securities with the South Carolina Tax Commission in compliance with the Act. Proof of such coverage shall be filed with the Engineer before work is started.
 - 3. If the Contractor fails to comply with the regulations of the South Carolina Tax Commission, two percent (2%) of each and every payment made to the Contractor shall be retained by the Owner to satisfy such requirements.

END OF SECTION.

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SECTION 01 70 00 - CONSTRUCTION SUPERVISION
(SUPERINTENDENT)

PART ONE - GENERAL

1-01 WORK INCLUDED:

- (A) The general contractor shall employ a full-time superintendent for supervision of construction and coordination of trades to maintain compliance with the contract documents.
- (B) Related Work: Documents affecting work of this Section include, but are not necessarily limited to General Conditions, Supplementary Conditions, and Sections in Division 00 of these specifications.

1-02 QUALITY ASSURANCE:

- (A) The general contractor's superintendent shall have not less than five (5) years direct experience in construction of the type and extent presented by this contract.
- (B) The superintendent shall have training and experience in reading contract documents and in interpreting all types of contract documents presented by this contract, architectural and engineering.

1-03 SUBMITTALS:

- (A) Upon request of architect, submit data demonstrating qualifications of person or persons the contractor proposes to engage as a construction superintendent. Such submittal is subject to approval by the architect.
- (B) Should the general contractor propose using multiple superintendents for specialty oversight or periods of limited duration; a submittal of intent, qualifications and job descriptions shall be mandatory for each superintendent proposed. Such submittal is subject to approval by the architect.
- (C) Rejection of any individual (s) for failure to meet the Quality standards established shall be the reserved right of the architect and such a decision shall be final.

1-04 RESPONSIBILITIES:

- (A) It is understood that the superintendent is the direct employee of the general contractor and the primary "charge of responsibilities," "job description": or "terms of employment" shall be established by the general contractor. However, the following responsibilities shall be considered as a minimum requirement for any person acting as construction superintendent under this contract.
 - (1) The superintendent shall be on the job site at all times during the forty (40) hour standard week as established by the contractor and/or shall be on the job site at any time a trade or subcontractor other than those forces directly

- employed by the general contractor are executing work under this contract.
- (2) The superintendent shall be familiar with the work in progress during all phases of construction, including the manpower on the project and the status of each trade's progress.
 - (3) All coordination between the owner's agents or normal operation in association with the construction site shall be the responsibility of the superintendent.
 - (4) The superintendent shall remain on the project from the Notice To Proceed throughout the punch list process, unless multiple superintendents have been approved as outlined under paragraph 1.04.

1-05 REPLACEMENT RESIGNATION OR TERMINATION:

- (A) Should the construction superintendent become ill, take vacation, or be absent from the project site for any extended period of time, the general contractor shall apprise the architect of the circumstances and duration of such absence, and provide a competent replacement for the period involved.
- (B) Should the construction superintendent resign or be terminated from the employment of the general contractor, the architect shall be apprized and qualifications of the proposed replacement superintendent shall be presented as indicated in 1.03 and 1.04 of this Section.

1-06 ARCHITECT'S AUTHORITY TO REPLACE:

- (A) The architect reserves the right to assess the performance of the superintendent for compliance with the Quality standards and minimum responsibilities as established in this Section. Should the superintendent fail to meet this criteria, in the opinion of the architect, the architect may require the general contractor to replace the superintendent.
- (B) Should such a replacement be requested, the general contractor shall present qualifications of the replacement superintendent in accordance with the provisions of this specification Section within seven (7) calendar days.

1-07 PENALTY FOR REMOVING OR REPLACING JOB SUPERINTENDENT:

Penalty for removing or replacing job superintendent if the superintendent is removed by the contractor for any other reason than noted above, the contractor shall be penalized liquidated damages in the amount of Five Hundred and No/100 (\$500.00) Dollars per calendar day for each day the contractor is not on site.

End of Section

SECTION 01 74 00 – CLEANING

PART ONE - GENERAL

1-01 DESCRIPTION

- A. Work included: Throughout the construction period, maintain the buildings and site in a standard of cleanliness as described in this section and the Contract Documents. Failure to do so may result in a reduction of the Contract Sum if the Program Manager has the required cleanup performed by others..
- B. Related work:
 - 1. Documents affecting work of this section include but are not necessarily limited to AIA A101 and A201, as amended, General Conditions, Supplemental Conditions, and Sections in Division 1 of these Specifications.
 - 2. In addition to standards described in this section, comply with requirements for cleaning as described in pertinent other sections of these Specifications.

1-02 QUALITY ASSURANCE

- A. Conduct daily cleaning and inspection and more often if necessary, to verify that requirements for cleanliness are being met.
- B. In addition to the standards described in this section, comply with pertinent requirements of governmental agencies having jurisdiction.

PART TWO - PRODUCTS

2-01 CLEANING MATERIALS AND EQUIPMENT

- A. Provide required personnel, equipment and materials needed to maintain the specified standard of cleanliness.

2-02 COMPATIBILITY

- A. Use only the cleaning materials and equipment which are compatible with the surface being cleaned, as recommended by the manufacturer of the material.

PART THREE - EXECUTION

3-01 PROGRESS CLEANING

A. General

1. Retain stored items in an orderly arrangement allowing maximum access, not impeding traffic or drainage and providing required protection of materials.
2. Do not allow accumulation of scrap, debris, waste material and other items not required for construction of this work.
3. At least twice each month and more often if necessary, completely remove all scrap, debris and waste material from the job site. Provide adequate storage for all items waiting removal from the job site, observing requirements for fire protection and protection of the ecology.
4. Locations on site for stored materials and/or debris must be pre- approved by the Architect and Owner.
5. Once concrete floor has been placed in any area, and overhead deck is in place, these areas shall be swept broom clean once per week or more often if necessary. When the building(s) is enclosed, dust control materials shall be used while sweeping to minimize dust accumulation on surfaces to receive finishes, ceiling grid, fixtures, mechanical systems, ductwork etc. The General Contractor will be solely responsible to assure clean systems and finishes throughout duration of project.

B. Site

1. Daily, and more often if necessary, inspect the site and pick up all scrap, debris and waste material. Remove such items to the place designated for their storage.
2. Weekly, and more often if necessary, inspect all arrangements of materials stored on the site. Restack, tidy, or otherwise service arrangements to meet the requirements of subparagraph 3.1.1A above.
3. Maintain the site in a neat, safe and orderly condition at all times.

3-02 FINAL CLEANING

- A. "Clean," for the purpose of this Article, and except as may be specifically provided otherwise, shall be interpreted as meaning the level of cleanliness generally provided by skilled cleaners using commercial quality building maintenance equipment and materials.
- B. Prior to completion of the work, remove from the job site all tools, surplus materials, equipment, scrap, debris, and waste. Conduct final progress cleaning as described in Article 3.1 above.
- C. Schedule final cleaning as approved by the Architect/Engineer to enable the Owner to accept a completely clean work site.

END OF SECTION

SECTION 01 77 00 - CLOSEOUT PROCEDURES

PART ONE - GENERAL

1-01 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 00 Specification Sections, apply to this Section.

1-02 SUMMARY

- B. This Section includes administrative and procedural requirements for contract closeout, including, but not limited to, the following:
 - 1. Inspection procedures.
 - 2. Project Record Documents.
 - 3. Operation and maintenance manuals.
 - 4. Warranties.
 - 5. Instruction of the Owner's personnel.
 - 6. Final cleaning.
- C. Specific requirements of each contract are also indicated in individual Specification Sections, All Bid Documents and on Drawings.

1-03 SUBSTANTIAL COMPLETION

- A. Preliminary Procedures: Before requesting inspection for determining date of Substantial Completion, complete the following. List items below that are incomplete in request.
 - 1. GC to prepare a list of items to be completed and corrected (punch list), the value of items on the list, and reasons why the Work is not complete.
 - 2. Advise Owner of pending insurance changeover requirements.
 - 3. Submit specific warranties, workmanship bonds, maintenance service agreements, final certifications, and similar documents.
 - 4. Obtain and submit releases permitting the Owner unrestricted use of the Work and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.
 - 5. Prepare and submit Project Record Documents, operation and maintenance manuals, Final Completion construction photographs, damage or settlement surveys, property surveys, and similar final record information.

6. Deliver tools, spare parts, extra materials, and similar items to location designated by the Owner. Label with manufacturer's name and model number where applicable.
 7. Terminate and remove temporary facilities from Project site, along with mockups, construction tools, and similar elements.
- B. Inspection: Submit a written request for inspection for Substantial Completion. On receipt of request, Architect & Owner will either proceed with inspection or notify Contractor of unfulfilled requirements. Architect will prepare the Certificate of Substantial Completion after inspection or will notify Contractor of items, either on Contractor's list or additional items identified by Architect & Owner, that must be completed or corrected before certificate will be issued.
1. Re-inspection: Request re-inspection when the Work identified in previous inspections as incomplete is completed or corrected.
 2. Results of completed inspection will form the basis of requirements for Final Completion.

1-04 FINAL COMPLETION

- C. Preliminary Procedures: Before requesting final inspection for determining date of Final Completion, complete the following:
1. Submit a final Application for Payment according to Division 1 Section "Payment Procedures."
 2. Submit certified copy of Architect's Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and dated by Architect. The certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.
 3. Submit evidence of final, continuing insurance coverage complying with insurance requirements.
 4. Final Completion must be achieved with 30 calendar days of Substantial Completion.
- D. Inspection: Submit a written request for final inspection for acceptance. On receipt of request, Architect & Owner will either proceed with inspection or notify Contractor of unfulfilled requirements. Architect will accept for review, a final Certificate for Payment after inspection or will notify Contractor of construction that must be completed or corrected before certificate will be issued.
1. Re-inspection: Request re-inspection when the Work identified in previous inspections as incomplete is completed or corrected.

1-05 LIST OF INCOMPLETE ITEMS (PUNCH LIST)

- E. Preparation: Submit three copies of list. Include name and identification of each space and area affected by construction operations for incomplete items and items needing correction including, if necessary, areas disturbed by Contractor that are outside the limits of construction.

- F. Organize list of spaces in sequential order.
 - 1. Organize items applying to each space by major element, including categories for ceiling, individual walls, floors, equipment, and building systems.
 - 2. Include the following information at the top of each page:
 - a. Project name.
 - b. Date.
 - c. Name of Architect.
 - d. Name of Contractor.
 - e. Page number.

1-06 PROJECT RECORD DOCUMENTS

- G. General: Do not use Project Record Documents for construction purposes. Protect Project Record Documents from deterioration and loss. Provide access to Project Record Documents for Architect & Owner, reference during normal working hours.
- H. Record Drawings: Maintain and submit one set of blue- or black-line white prints of Contract Drawings and Shop Drawings.
 - 1. Mark Record Prints to show the actual installation where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to prepare the marked-up Record Prints.
 - a. Give particular attention to information on concealed elements that cannot be readily identified and recorded later.
 - b. Accurately record information in an understandable drawing technique.
 - c. Record data as soon as possible after obtaining it. Record and check the markup before enclosing concealed installations.
 - d. Mark Contract Drawings or Shop Drawings, whichever is most capable of showing actual physical conditions, completely and accurately. Where Shop Drawings are marked, show cross-reference on Contract Drawings.
 - 2. Mark record sets with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at the same location.
 - 3. Mark important additional information that was either shown schematically or omitted from original Drawings.
 - 4. Note Construction Change Directive numbers, Change Order numbers, alternate numbers, and similar identification where applicable.
 - 5. Identify and date each Record Drawing; include the designation "PROJECT RECORD DRAWING" in a prominent location. Organize into manageable sets; bind each set with durable paper cover sheets. Include identification on cover sheets.

6. Provide a spreadsheet inventory list of all filters, sizes, locations, etc. prior to final completion / payment.
- I. Record Specifications: Submit one copy of Project's Specifications, including addenda and contract modifications. Mark copy to indicate the actual product installation where installation varies from that indicated in Specifications, addenda, and contract modifications.
 1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
 2. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.
 3. Note related Change Orders, Record Drawings, and Product Data, where applicable.
- J. Record Product Data: Submit one copy of each Product Data submittal. Mark one set to indicate the actual product installation where installation varies substantially from that indicated in Product Data.
 1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
 2. Include significant changes in the product delivered to Project site and changes in manufacturer's written instructions for installation.
 3. Note related Change Orders, Record Drawings, and Record Specifications, where applicable.
- K. Miscellaneous Record Submittals: Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference.

1-07 WARRANTIES

- L. Submittal Time: Submit written warranties on request of Architect for designated portions of the Work where commencement of warranties other than date of Substantial Completion is indicated.
- M. Partial Occupancy: Submit properly executed warranties within 15 working days of completion of designated portions of the Work that are completed and occupied or used by The Presbyterian Communities during construction period.
- N. Organize warranty documents into an orderly sequence based on the table of contents of the Project Manual.
 1. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2-by-11-inch paper.
 2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify the product or installation. Provide a typed

description of the product or installation, including the name of the product and the name, address, and telephone number of Installer.

3. Identify each binder on the front and spine with the typed or printed title "WARRANTIES," Project name, and name of Contractor.

- O. Provide additional copies of each warranty to include in operation and maintenance manuals.
- P. Provide 3 copies of all closeout documents to Architect for distribution to Owner. This includes O & M Manuals, Project Documents, Approvals, Certificates, and all warranty information.

1-08 MAINTENANCE TRAINING/EQUIPMENT DEMONSTRATION VIDEOS

- A. Equipment Demonstration Videos: The contractor shall have each equipment demonstration videotaped by the factory representative for all demonstrations provided at the school. The demonstrations shall be videoed during the actual presentation class at the site with the maintenance staff and contractor present. Video shall be easily viewed and set up on tripod for smooth picture quality.

Number of videos provided for each demonstration - two (2) copies.

The following video demonstrations are required as noted below.

1. Complete demonstration of the equipment and proper use of the HVAC System by the factory representative. This needs to be a comprehensive demonstration of the system itself, operational instructions, routine maintenance items, and emergency troubleshooting tips.
2. Complete demonstration of the Fire Alarm panel and system as well as working with troubles, etc.
3. Complete demonstration of any Fire Curtains and the resetting of such curtains.
4. Complete demonstration of the Electrical Switchgear as well as demonstration on adjustments that need to be made when power fluctuates.

PART TWO - PRODUCTS

2-01 MATERIALS

- Q. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

PART THREE – EXECUTION

3-01 FINAL CLEANING

- A. General: Provide final cleaning. Final Cleaning is the responsibility of the General Contractor. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.
- B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.
 - 1. Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:
 - a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
 - b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
 - c. Pressure-wash all concrete and paved surfaces.
 - d. Rake grounds that are neither planted nor paved to a smooth, even-textured surface.
 - e. Remove tools, construction equipment, machinery, and surplus material from Project site.
 - f. Remove snow and ice to provide for safe access to facility. Continue providing this service until substantial completion has been achieved.
 - g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults / closets, manholes, attics, mezzanines and similar spaces.
 - h. Sweep concrete floors broom clean with sweeping compound.
 - i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; have carpets professionally shampooed / cleaned if visible soil or stains remain.
 - j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and other noticeable, vision obscuring materials. Replace chipped or broken glass and other materials. Polish mirrors and glass, taking care not to scratch surfaces.
 - k. Remove labels that are not permanent.
 - l. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.
 - m. Remove any paint from "UL" and similar labels, including mechanical and electrical nameplates.

- n. Wipe surfaces of mechanical and electrical equipment, elevator equipment and similar equipment. Remove excess lubrication, paint and mortar drippings, and other foreign substances.
 - o. Replace parts subject to unusual operating conditions.
 - p. Clean plumbing fixtures to a sanitary condition, free of stains, including stains from water exposure.
 - q. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of all diffusers, registers and grilles.
 - r. Clean ducts, blowers and coils.
 - s. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned out bulbs, and those noticeably dimmed by hours of use, and defective and / or noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.
 - t. GC to be responsible for professional final cleaning of the interior of the building to include 3 coats of wax on all VCT.
 - u. All fixtures to be wiped clean. Food Service equipment to be cleaned. The GC warrants that it will provide the facility in a "white glove test" condition to the Owner.
 - v. GC to be responsible for final cleaning of building exterior to include windows and all horizontal and vertical surfaces.
 - w. GC to pressure-wash all exterior hard surfaces.
 - x. GC to provide the same cleaning as listed herein for adjacent spaces that have been affected by the construction process.
- C. Comply with safety standards for cleaning. Do not burn waste materials. Do not bury debris or excess materials on the Owner's property. Do not discharge volatile, harmful, or dangerous materials into drainage systems. Remove waste materials from Project site and dispose of lawfully.
- D. Pest Control: Engage an experience, licensed exterminator to make a final inspection and rid Project of any rodents, insects, and other pests. Exterminator to provide a written report to Owner prior to Substantial Completion being awarded.

END OF SECTION

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SECTION 01 78 30 - WARRANTIES AND BONDS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and other Division-00 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section specifies general administrative and procedural requirements for warranties and bonds required by the Contract Documents, including manufacturers' standard warranties on products and special warranties.
 - 1. Refer to the General Conditions for terms of the Contractor's special warranty of workmanship and materials.
 - 2. General closeout requirements are included in Section "Project Closeout."
 - 3. Specific requirements for warranties for the Work and products and installations that are specified to be warranted, are included in the individual Sections of Divisions-2 through -28.
 - 4. Certifications and other commitments and agreements for continuing services to Owner are specified elsewhere in the Contract Documents.
- B. Disclaimers and Limitations: Manufacturer's disclaimers and limitations on product warranties do not relieve the Contractor of the warranty on the Work that incorporates the products, nor does it relieve suppliers, manufacturers, and subcontractors required to countersign special warranties with the Contractor.
- C. Separate Prime Contracts: Each prime Contractor is responsible for warranties related to its own Contract.

1.3 DEFINITIONS

- A. Standard Product Warranties are preprinted written warranties published by individual manufacturers for particular products and are specifically endorsed by the manufacturer to the Owner.
- B. Special Warranties are written warranties required by or incorporated in the Contract Documents, either to extend time limits provided by standard warranties or to provide greater rights for the Owner.

1.4 WARRANTY REQUIREMENTS

- A. Related Damages and Losses: When correcting warranted Work that has failed, remove and replace other Work that has been damaged as a result of such failure or that must be removed and replaced to provide access for correction of warranted Work.
- B. Reinstatement of Warranty: When Work covered by a warranty has failed and been corrected by replacement or rebuilding, reinstate the warranty by written endorsement. The reinstated warranty shall be equal to the original warranty with an equitable adjustment for depreciation.
- C. Replacement Cost: Upon determination that Work covered by a warranty has failed, replace or rebuild the Work to an acceptable condition complying with requirements of Contract Documents. The Contractor is responsible for the cost of replacing or rebuilding defective Work regardless of whether the Owner has benefited from use of the Work through a portion of its anticipated useful service life.
- D. Owner's Recourse: Written warranties made to the Owner are in addition to implied warranties, and shall not limit the duties, obligations, rights and remedies otherwise available under the law, nor shall warranty periods be interpreted as limitations on time in which the Owner can enforce such other duties, obligations, rights, or remedies.
 - 1. Rejection of Warranties: The Owner reserves the right to reject warranties and to limit selections to products with warranties not in conflict with requirements of the Contract Documents.
- E. The Owner reserves the right to refuse to accept Work for the Project where a special warranty, certification, or similar commitment is required on such Work or part of the Work, until evidence is presented that entities required to countersign such commitments are willing to do so.

1.5 SUBMITTALS

- A. Submit written warranties to the Architect prior to the date certified for Substantial Completion. If the Architect's Certificate of Substantial Completion designates a commencement date for warranties other than the date of Substantial Completion for the Work, or a designated portion of the Work, submit written warranties upon request of the Architect.
 - 1. When a designated portion of the Work is completed and occupied or used by the Owner, by separate agreement with the Contractor during the construction period, submit properly executed warranties to the Architect within fifteen days of completion of that designated portion of the Work.
- B. When a special warranty is required to be executed by the Contractor, or the Contractor and a subcontractor, supplier or manufacturer, prepare a written

document that contains appropriate terms and identification, ready for execution by the required parties. Submit a draft to the Owner through the Architect for approval prior to final execution.

- C. Forms for special warranties are included at the end of this Section. Prepare a written document utilizing the appropriate form, ready for execution by the Contractor, or the Contractor and subcontractor, supplier or manufacturer. Submit a draft to the Owner through the Architect for approval prior to final execution.
 - 1. Refer to individual Sections of Divisions-2 through -28 for specific content requirements, and particular requirements for submittal of special warranties.
- D. Form of Submittal: At Final Completion compile two copies of each required warranty and bond properly executed by the Contractor, or by the Contractor, subcontractor, supplier, or manufacturer. Organize the warranty documents into an orderly sequence based on the table of contents of the Project Manual.
- E. Bind warranties and bonds in heavy-duty, commercial quality, durable 3-ring vinyl covered loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2" by 11" paper.
 - 1. Provide heavy paper dividers with celluloid covered tabs for each separate warranty. Mark the tab to identify the product or installation. Provide a typed description of the product or installation, including the name of the product, and the name, address and telephone number of the installer.
 - 2. Identify each binder on the front and the spine with the typed or printed title "WARRANTIES AND BONDS, the Project title or name, and the name of the Contractor.
 - 3. When operating and maintenance manuals are required for warranted construction, provide additional copies of each required warranty, as necessary, for inclusion in each required manual.

PART 2 - PRODUCTS (Not Applicable).

PART 3 - EXECUTION

3.1 SCHEDULE OF WARRANTIES

- A. Schedule: Provide warranties and bonds on products and installations as specified in the following Sections:

Water Proofing: Section 07 13 00 – Waterproofing

Exterior Insulation and Finish System: Section 07 24 00 - EIFS

Asphalt Shingles: Section 07 31 13 – Asphalt Shingles

Flush Wood Doors: Section 08 14 16 - Wood Doors
Aluminum – Framed Storefront: Section 08 41 13 – Doors and Frame
Wood Windows Windows: Section 08 61 00 – Vinyl Clad Wood Windows
Hardware: Section 08 71 00 – Door Hardware
Insulating Glass: Section 08 80 00 - Glass and Glazing
Sheet Vinyl Flooring: Section 09 66 60 – Sheet Vinyl Flooring
Food Service Equipment: Section 11 40 00 - Food Service Equipment
Residential Appliances: Section 11 30 00 – Residential Appliances
Water Heaters: Section 22 01 00 – Mechanical; Plumbing
Packaged Heating and Cooling: Section 23 01 00 – HVAC; Mechanical
Units: Heating and Cooling Units
Termite Control Treatment: Section 31 31 16 – Termite Treatment

END OF SECTION

DIVISION NO. 02 – EXISTING CONDITIONS

SECTION 02 41 19 - SELECTIVE DEMOLITION

PART ONE - GENERAL

1-01 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1-02 SUMMARY

- A. This Section requires the selective removal and subsequent offsite disposal of the following:
 - 1. Removal of the existing poured urethane gym floor.
 - 2. Removal of the existing door frames and doors as required.
 - 3. Removal of the existing bleachers.
- B. Related work specified elsewhere:
 - 1. Remodeling construction work and patching are included within the respective sections of specifications, including removal of materials for reuse and incorporation into remodeling or new construction.
 - 2. Relocation of pipes, conduits, ducts, and other mechanical and electrical work is specified in other Divisions.

1-03 SUBMITTALS

- A. General: Submit the following in accordance with Conditions of Contract and Divisions 00 and 01 Specification Sections.
- B. Schedule indicating proposed sequence of operations for selective demolition work to Owner's Representative for review prior to start of work. Include coordination for shutoff, capping, and continuation of utility services as required, together with details for dust and noise control protection.
 - 1. Provide detailed sequence of demolition and removal work to ensure uninterrupted progress of Owner's on-site operations.
 - 2. Coordinate with Owner's continuing occupation of portions of existing building and with Owner's partial occupancy of completed new addition.
- C. Photographs of existing conditions of structure surfaces, equipment, and adjacent improvements that might be misconstrued as damage related to

removal operations. File with Owner's Representative prior to start of work.

1-04 JOB CONDITIONS

- A. Occupancy: Owner will occupy portions of the building immediately adjacent to areas of selective demolition. Conduct selective demolition work in manner that will minimize need for disruption of Owner's normal operations. Provide minimum of 48 hours advance notice to Owner of demolition activities that will affect Owner's normal operations.
- B. Condition of Structures: Owner assumes no responsibility for actual condition of items or structures to be demolished.
 - 1. Conditions existing at time of inspection for bidding purposes will be maintained by Owner insofar as practicable. However, minor variations within structure may occur by Owner's removal and salvage operations prior to start of selective demolition work.
- C. Partial Demolition and Removal: Items indicated to be removed but of salvageable value to Contractor may be removed from structure as work progresses. Transport salvaged items from site as they are removed.
 - 1. Storage or sale of removed items on site will not be permitted.
- C. Protections: Provide temporary barricades and other forms of protection to protect Owner's personnel and general public from injury due to selective demolition work.
 - 1. Provide protective measures as required to provide free and safe passage of Owner's personnel and general public to occupied portions of building.
 - 2. Erect temporary covered passageways as required by authorities having jurisdiction.
 - 3. Provide interior and exterior shoring, bracing, or support to prevent movement, settlement, or collapse of structure or element to be demolished and adjacent facilities or work to remain.
 - 4. Protect from damage existing finish work that is to remain in place and becomes exposed during demolition operations.
 - 5. Protect floors with suitable coverings when necessary.
 - 6. Construct temporary insulated dustproof partitions where required to separate areas where noisy or extensive dirt or dust operations are performed. Equip partitions with dustproof doors and security locks.
 - 7. Provide temporary weather protection during interval between demolition and removal of existing construction on exterior surfaces and installation of new construction to ensure that no water leakage or damage occurs to structure or interior areas of existing building.
 - 8. Remove protections at completion of work.
- D. Damages: Promptly repair damages caused to adjacent areas by demolition work.

- E. Traffic: Conduct selective demolition operations and debris removal to ensure minimum interference with roads, streets, walks, and other adjacent occupied or used facilities.
 - 1. Do not close, block, or otherwise obstruct streets, walks, or other occupied or used facilities without written permission from authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways if required by governing regulations.
- F. Flame Cutting: Do not use cutting torches for removal until work area is cleared of flammable materials. At concealed spaces, such as interior of ducts and pipe spaces, verify condition of hidden space before starting flame-cutting operations. Maintain portable fire suppression devices during flame-cutting operations.
- G. Utility Services: Maintain existing utilities indicated to remain in service and protect them against damage during demolition operations.
 - 1. Do not interrupt utilities serving occupied or used facilities, except when authorized in writing by authorities having jurisdiction. Provide temporary services during interruptions to existing utilities, as acceptable to governing authorities.
 - 2. Maintain fire protection services during selective demolition operations.
- H. Environmental Controls: Use water sprinkling, temporary enclosures, and other methods to limit dust and dirt migration. Comply with governing regulations pertaining to environmental protection.
 - 1. Do not use water when it may create hazardous or objectionable conditions such as ice, flooding, and pollution.

PART TWO - PRODUCTS (Not Applicable)

PART THREE - EXECUTION

3-01 PREPARATION

- A. General: Provide interior and exterior shoring, bracing, or support to prevent movement, settlement, or collapse of areas to be demolished and adjacent facilities to remain.
 - 1. Cease operations and notify Owner's Representative immediately if safety of structure appears to be endangered. Take precautions to support structure until determination is made for continuing operations.
 - 2. Cover and protect furniture, equipment, and fixtures from soilage or damage when demolition work is performed in areas where such items have not been removed.
 - 3. Erect and maintain dust-proof partitions and closures as required to prevent spread of dust or fumes to occupied portions of the building.
 - a. Where selective demolition occurs immediately adjacent to

- occupied portions of the building, construct dust-proof partitions of minimum 4-inch studs, 5/8-inch drywall (joints taped) on occupied side, 1/2-inch fire-retardant plywood on demolition side. Fill partition cavity with sound-deadening insulation.
- b. Provide weatherproof closures for exterior openings resulting from demolition work.
4. Locate, identify, stub off, and disconnect utility services that are not indicated to remain.
- a. Provide bypass connections as necessary to maintain continuity of service to occupied areas of building. Provide minimum of 48 hours advance notice to Owner if shutdown of service is necessary during changeover.

3-02 DEMOLITION

- A. General: Perform selective demolition work in a systematic manner. Use such methods as required to complete work indicated on Drawings in accordance with demolition schedule and governing regulations.
- 1. Demolish concrete and masonry in small sections. Cut concrete and masonry at junctures with construction to remain using power-driven masonry saw or hand tools; do not use power-driven impact tools.
 - 2. Locate demolition equipment throughout structure and promptly remove debris to avoid imposing excessive loads on supporting walls, floors, or framing.
 - 3. Provide services for effective air and water pollution controls as required by local authorities having jurisdiction.
 - 4. Demolish foundation walls to a depth of not less than 12 inches below existing ground surface. Demolish and remove below-grade wood or metal construction. Break up below-grade concrete slabs.
 - 5. For interior slabs on grade, use removal methods that will not crack or structurally disturb adjacent slabs or partitions. Use power saw where possible.
 - 6. Completely fill below-grade areas and voids resulting from demolition work. Provide fill consisting of approved earth, gravel, or sand, free of trash and debris, stones over 6 inches in diameter, roots, or other organic matter.
- B. If unanticipated mechanical, electrical, or structural elements that conflict with intended function or design are encountered, investigate and measure both nature and extent of the conflict. Submit report to Owner's Representative in written, accurate detail. Pending receipt of directive from Owner's Representative, rearrange selective demolition schedule as necessary to continue overall job progress without undue delay.

3-03 SALVAGED MATERIALS / RECYCLED MATERIAL

- A. Salvaged Items: Where indicated on Drawings as "Salvage - Deliver to Owner," carefully remove indicated items, clean, store, and turn over to Owner

and obtain receipt.

1. Historic artifacts, including cornerstones and their contents, commemorative plaques and tablets, antiques, and other articles of historic significance, remain property of Owner. Notify Owner's Representative if such items are encountered and obtain acceptance regarding method of removal and salvage for Owner.
2. Contractor shall recycle any existing ceiling tile removed for construction. The contractor shall contact Armstrong Ceiling at (877)276-7876 for acceptance of material.

3-04 DISPOSAL OF DEMOLISHED MATERIALS

- A. Remove from building site debris, rubbish, and other materials resulting from demolition operations. Transport and legally dispose off site.
 1. If hazardous materials are encountered during demolition operations, comply with applicable regulations, laws, and ordinances concerning removal, handling, and protection against exposure or environmental pollution.
 2. Burning of removed materials is not permitted on project site.

3-05 CLEANUP AND REPAIR

- A. General: Upon completion of demolition work, remove tools, equipment, and demolished materials from site. Remove protections and leave interior areas broom clean.
 1. Repair demolition performed in excess of that required. Return elements of construction and surfaces to remain to condition existing prior to start operations. Repair adjacent construction or surfaces soiled or damaged by selective demolition work.

END OF SECTION

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DIVISION NO. 07 – THERMAL & MOISTURE PROTECTION

SECTION 07 10 00 - WATERPROOFING AND DAMPPROOFING

PART 1 GENERAL

- 1.01 REQUIREMENTS:** The requirements of all sections of Division One apply to work under this section.
- 1.02 WORK INCLUDED:** Furnish labor and materials necessary to complete Vapor retarding under wood flooring as required by Athletic Wood Supplier.

PART 2 PRODUCTS

- 2.01 MATERIALS:**
- (A) Vapor Barrier: 6 MIL MOISTOP with joints taped.

PART 3 EXECUTION

- 3.01 VAPOR BARRIER:** Apply 6 mil MOISTOP 4 ply vapor barrier over surface of stone aggregate ASTM #5 compacted subgrade before placing 2" of coarse sand. Lap joints not less than 6". Protect from damage and if punctured, seal punctures with a patch before concrete is poured. Take proper precautions to install properly. See Specification Section 03300 (Concrete) for additional information. Tape all horizontal and vertical joints prior to installation of Wood Subfloor material.

PART 4 WARRANTIES

- 4.01 MEMBRANES:** All membranes to be guaranteed watertight for a minimum of twenty (20) years.
- 4.02 INSTALLATION:** The entire installation shall be guaranteed watertight.
- 4.03 PRODUCTS:** All products used shall be either from or approved by the same manufacturer.

END OF SECTION

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DIVISION NO. 08 – FINISHES

SECTION 08 11 00 - STANDARD STEEL DOORS AND FRAMES

PART 1 GENERAL

- 1.01 REQUIREMENTS:** The requirements of all sections of Division One apply to work under this section.
- 1.02 WORK INCLUDED:** Furnish labor and materials necessary to complete hollow metal doors and frames indicated, specified or both. Comply with ANSI/SDI-100, SDI-150, and NFPA #80.
- 1.03 SHOP DRAWINGS:** The Contractor shall submit six (6) copies of Checked shop drawings for approval by the Architect.

PART 2 PRODUCTS AND EXECUTION

2.01 HOLLOW METAL DOORS

- (A) Doors shall be CECO. Approved Equals: D&D Specialties, Curries Company, Palmetto Metal Products, and Advance Door/Republic.
- (B) Doors shall be made of 16 gauge roller leveled prime quality cold rolled steel sheets. Vertical stiffeners shall be No. 16 gauge channel shaped steel on nominal 6" spaces, welded to both face sheets on 5" centers.
- (C) Top and bottom channels shall be No. 14 gauge. The Vertical edge channels shall be No. 14 gauge. The vertical edge joint between the face sheets shall be continuously welded together and to the vertical edge channels. This joint shall be ground smooth to provide a complete flush door with no exposed seams.
- (D) All doors shall be filled with spun material insulation.
- (E) Doors with glass light opening shall have the openings formed into the face sheets so that no glass frame is required. Glazing beads shall be recessed behind the formed openings in the face sheets beveled, and attached without screws. Glazing bead corners shall present a mitered appearance.
- (F) The tops of all exterior doors shall be closed flush to prevent accumulation of moisture and rain water.
- (G) If rated doors are indicated, provide labeled door and frames.
- (H) Doors shall be bonderized and coated with one prime coat of rust inhibitive paint.

2.02 METAL DOORS AND WINDOW FRAMES

- (A) Frames shall be custom fabricated units, full wrap around, as fabricated by Lockhart Manufacturing Company of Charlotte, North Carolina, or approved equal.
- (B) Frames for units including 3'0" wide and less shall be fabricated from 14 ga. material, frames for units over 3'0" wide shall be fabricated from 15 ga. material.
- (C) Frames shall be constructed in accordance with the profile details as required. All corners shall be shop assembled, weld corners, molds, returns to insure a rigid joint.
- (D) Furnish not less than 3 adjustable anchors per jamb and a floor anchor at each jamb. Furnish a removable steel angle spreader with each frame to insure rigidity during transit and handling.

- (E) Frames shall be mortised and reinforced with steel plate and tapped for hardware, templates will be furnished. Surface type hardware such as closures, holders, etc., will be reinforced for field drilling and tapping. Mortar boxes are to be installed over all hardware cut-outs.
- (F) Provide labeled frames where scheduled.
- (G) Frames shall be bonderized and flow coated with one prime coat of rust inhibitive primer.
- (H) No hammer marks will be accepted.
- (I) At certain doors on the Door Schedule, manufacturer shall install at factory in frame a backer box to provide space for $\frac{3}{4}$ " electrical conduit for card reader security. Coordinate with electrical contractor location and size required. This occurs at approximately five (5) areas around the school. See Door Schedule for number of doors affected.

SPECIAL NOTE: Hollow metal frames for 16" thick masonry walls where they occur may be 12" butt frames at the gymnasium only. All hollow metal frame center mullions shall be filled solid with grout or concrete or as recommended by frame manufacturer for a stiff, solid, secure frame.

2.03 ALUMINUM DOOR AND FRAME

Provide where shown in gym locker/shower area Aluminum doors and frames. Doors shall be md240i as manufactured by Alutech Corp. or approved equal. Doors shall have a face sheet of on piece .040 smooth aluminum of anodizing quality, with a minimum tensile strength of 22,000 psi. Face sheets shall be concealed under a $\frac{1}{2}$ " edge cap full perimeter.

All voids between the sub-frames shall be filled with a isocyanurate rigid foam board that is fire retardant and a R-value of 9. Core shall be laminated between two sheets of $\frac{1}{8}$ " tempered hardboard for impact reinforcing. Hardboard shall be same size as face sheets. Sub-frame, face sheets and insulated core shall be bonded together with a thermosetting adhesive under pressure to form a water, heat and chemical resistant bond.

The sub-frame shall be $1 \frac{3}{8}$ " x $4 \frac{1}{2}$ " 6063T5 tubular aluminum sub-frame with a $1 \frac{3}{4}$ " integral edge cap at hinge and lock side with an applied edge cap top and bottom to insure proper protection from water.

Door finish shall be clear anodized with a minimum coating of .4 mils.

Doors shall be prepared for a 1-1/2 pair of template type hinges ($4 \frac{1}{2}$ " x $4 \frac{1}{2}$ " x .134) one cylindrical deadbolt $2 \frac{3}{4}$ " backset $2 \frac{1}{8}$ " bore and push pull hardware. Hinges and strike shall be $\frac{1}{4}$ " aluminum plate.

Frame shall be FR450S as manufactured by Alutech Corp. or approved equal. Frame shall be a 6063-T5 extruded aluminum alloy with nominal wall thickness of .090". All screws and fasteners shall be stainless steel. Weather-stripping shall be Santoprene Thermoplastic black rubber. Finish shall be clear anodized 204R1 with a minimum thickness of .4 mils.

END OF SECTION

SECTION 08 14 00 - WOOD DOORS

PART 1 GENERAL

- 1.01 REQUIREMENTS:** The requirements of all sections of Division One apply to work under this section. All plastic laminate doors shall have a lifetime warranty against swelling, delaminating and warping.
- 1.02 WORK INCLUDED:** Furnish plant labor and materials to complete wood doors indicated, specified or both.
- 1.03 WORK NOT INCLUDED:** Labor for installing doors is included in Section 6 -CARPENTRY.
- 1.04 PROTECTION:** Protect doors from moisture and damage before and during delivery. Do not install until after concrete and adjacent masonry work has been completed and made dry. Store wood doors flat on level surface in a dry, ventilated structure. Doors to be delivered to job site only after HVAC system is operational.
- 1.05 FABRICATION:** Confirm requirements for tolerances, undercuts, grilles, lights, hardware and similar items prior to fabrication.
- 1.06 PANELED, LOUVERED, ETC., DOORS:** In general conform to the requirements of the NWMA. Lumber for these doors shall be manufacturer's standard grade and species as approved by the Architect.
- 1.07 SHOP DRAWINGS:** Submit shop drawings and qualification documents on all doors.
- 1.08 RATED DOORS:** Where rated doors are indicated, provide label on same. Match grade and face veneers of non-rated doors.

PART 2 PRODUCTS

- 2.01 FLUSH DOORS:** Doors shall be 1-3/4" solid stave lumber core doors with plain sliced stain grade red oak faces as manufactured by Algoma, VT Industries, Mohawk, or approved equal. All doors shall have pre-finished stain on all sides. Color to be selected by owner.
- 2.02 SPECIAL NOTES:**
- (A) This subcontractor shall refer to the mechanical plans and is responsible for coordinating with the mechanical subcontractor regarding the above work, including louver location and installation of same in all wood doors.
- (B) For undercut doors and louvers; it shall be the responsibility of this subcontractor to undercut all doors as indicated on the mechanical plans and also to cut all openings in all doors for louvers of the size and shape as indicated and shown on the mechanical plans. The mechanical subcontractor shall be responsible for furnishing and installing these louvers for the openings. NO louvers shall be installed in rated doors.

End of Section

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DIVISION NO. 09 – FINISHES

SECTION 09 64 44 - (ATHLETIC) WOOD FLOORING

PART 1 GENERAL

1.01 REQUIREMENTS: The requirements of all sections of Division One apply to work under this section.

1.02 WORK INCLUDED: Furnish labor and materials necessary to complete the maple flooring indicated, specified or both.

- (A) This document specifies a wood strip gymnasium floor system consisting, in general, of wood sub-flooring, maple flooring, vapor barrier, sanding, sealers, finishes, gamelines and wall base.
- (B) The general contractor shall provide a concrete slab smooth troweled and level to a tolerance of 1/8" in a 10'0" radius, subject to the approval of the wood flooring contractor. LABOR AND MATERIALS NECESSARY TO PUT THE CONCRETE SLAB IN ACCEPTABLE CONDITION (HIGH AREAS GROUND DOWN AND LOW AREAS FILLED WITH APPROVED LEVELING COMPOUNDS) SHALL BE THE RESPONSIBILITY OF THE GENERAL CONTRACTOR.
 - 1. Slab Depression: Slab is existing and only recessed 1/4" +/- for existing poured urethane floor (to be removed)
 - 2. Concrete subfloors shall have an adequate moisture barrier beneath and at the perimeter of the slab.

1.03 QUALITY ASSURANCE

- (A) Supplier Qualifications
 - 1. Supplier of wood flooring shall be Action Flooring Systems.
- (B) Installer Qualifications
 - 1. Flooring contractor shall be a firm experienced in maple flooring systems.
- (C) Flooring shall be stored on the premises a minimum of seven (7) days before installation commences, or as required for acclimation. Final determination as to acclimation will be made by the flooring contractor.
- (D) All flooring bundles should be broken and loosely piled to acclimate the flooring to moisture conditions in the building.

1.04 WORKING CONDITIONS

- (A) The floor system shall not be delivered or installed until all masonry, plastering, tile work and all overhead mechanical trades are completed and building is enclosed and weathertight.
- (B) Permanent heat, light and ventilation shall be installed and operating during and after installation, maintaining a temperature range of 55° F to 75° F and a relative humidity range of 35% to 50%.

1.05 WARRANTY

- (A) Action Floor Systems hereby warrants the ANCHOR_FLEX flooring materials to be free from manufacturing defects for a period of one year from the date of substantial completion.

PART 2 PRODUCTS

2.01 MATERIALS

- (A) Flooring
 - 1. Flooring shall be ACTION Floor Systems ANCHOR FLEX Floor System, Maple, 25/32" thick 2-1/4" wide Action Hard Maple Flooring, First Grade, continuous tongue and groove and end-matched MFMA Northern Hard Maple, grade marked and stamped as manufactured by Action Floor Systems. OPTION: Maple Flooring shall be treated with WOODLIFE F preservative.
- (B) Vapor Barrier shall be 6 mil polyethylene with taped joints.
- (C) Subfloor
 - 1. Thrust-A-Cushion pads shall be 3/8" thick, 2-1/4" x 3".
 - 2. Panels shall be 23/32" x 2' x 8' APA-rated sheathing plywood, exposure 1. Minimum APA Span rating of 48/24.
- (D) Wall Base shall be 3" x 4" Gym Cove heavy duty molded, vented rubber base with premolded outside corners as supplied by Action Floor System. Specify color: Black.
- (E) Finish material shall be as specified by architect selected from the latest listing provided by the MFMA and shall be applied according to manufacturer's instructions.
- (F) Flooring fasteners shall be as per manufacturer's requirements.

PART 3 EXECUTION

3.01 INSPECTION

- (A) Inspect concrete slab for proper tolerance and dryness, and report any discrepancies to the general contractor for correction.
- (B) The concrete slab shall be cleaned of all debris so flooring contractor will have adequate access to work surface.

3.02 INSTALLATION

- (A) Cover concrete with 6 mil polyethylene, sealing and lapping joints a minimum of 4".
- (B) Install first layer of 25/32" x 2' x 8' sheathing grade plywood with 5/8" foam cushion pads attached. Provide expansion void along perimeter and all vertical obstructions. Install perimeter rows of cushions with not more than 6" from edge of sheet to center of cushion. Install plywood per manufacturer's recommendation.
- (C) Install finish flooring parallel with main playing court by power nailing approximately 12" o.c.
 - 1. Space joints between flooring strips to allow for intermediate expansion, as designated by the installer and in accordance with humidity conditions in the region.
 - 2. Provide 2" expansion voids at the perimeter and at all vertical obstructions.

3.03 FLOOR SANDING

- (A) Machine sand with coarse, medium and fine paper to a smooth, even and uniform surface.
- (B) Remove sanding dust from entire surface by tack or vacuum.
- (C) Refer to MFMA sanding and finishing guide for procedures.

3.04 FINISHING

- (A) Inspect entire area of floor to insure that surface is acceptable for finishing, completely free from sanding dust and perfectly clean.
- (B) Apply seal and finish per manufacturer's instructions.
- (C) Apply two (2) coats of seal and two (2) coats of finish.
- (D) Buff and clean floor between each coat.
- (E) Paint game lines as shown on drawings, between seal and first coat of finish.
- (F) Game line paint shall be compatible with finish.

3.05 BASE INSTALLATION

- (A) Install Gym Cove base, mitering inside corners, and anchoring to walls with base cement or screws and anchors.

3.06 MAINTENANCE

- (A) Upon completion of floor installation, the owner, attendants or individuals in charge and responsible for the upkeep of the building are to see that the care and maintenance instructions of the Horner or MFMA Care Card are followed.

END OF SECTION

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SECTION 09 91 00 - PAINTING

PART 1 GENERAL

1.01 WORK INCLUDED

- (A) Furnish labor and materials to complete painting and finishing work indicated, specified or both.
- (B) Except as otherwise specified, paint and/or finish all exposed surfaces of wood, plaster, metal, concrete masonry units, stucco, concrete, unfinished metals, structural steel, gypsum wallboard and cementitious fiber boards or other materials as required to make a complete job.
- (C) The contractor shall examine the specifications for the various other trades and shall thoroughly familiarize himself with all provisions regarding their painting and shall understand that all surfaces that are left unfinished by the requirements of the other sections shall be painted under this section. The contractor shall understand that all work specified under this section shall be in addition to shop & mill coats, priming & field coats specified in other sections.
- (D) The contractor shall do all touching up of shop coats and field coats of paint on structural steel and miscellaneous steel or iron as required and/or specified.

1.02 WORK NOT INCLUDED

- (A) Items and surfaces that do NOT require painting shall include:
 - 1. Prefinished items: Casework, doors, elevator entrances and cabs, prefabricated metal building, equipment and similar items specified under other sections.
 - 2. Concealed surfaces: Inside dumbwaiter, elevator and duct shafts, pipe basements, crawl spaces, pipe tunnels, above ceilings, attics, except as otherwise specified. Surfaces concealed behind permanently installed casework and equipment. Structural steel encased in concrete, fire rating enclosures or in spray fireproofing.
 - 3. Finished surfaces: Anodized aluminum, stainless steel, chromium plating, copper, brass, except as otherwise specified.
 - 4. Moving and operating parts: Shafts, chains, gears, mechanical and electrical parts such as valve stems, operators, linkages, sprinkler heads, sensing devices.
 - 5. Labels: Any code required label such as Underwriters Laboratories Inc., or Factory Mutual Research Corporation, identification plates, instruction plates, performance rating, nomenclature.
 - 6. Hot-dip galvanized metal, except where specifically specified to be painted. Exterior galvanized metal such as chain link fence and gates, corrugated metal, gratings. Metal safety treads and nosings.
 - 7. Concrete curbs, gutters, pavements, retaining walls, and foundations.
 - 8. Face brick.

1.03 QUALITY ASSURANCE

- (A) Applicator Qualifications: Engage an experienced applicator who has completed painting system applications similar in material and extent to those indicated for the Project that have resulted in a construction period of successful in-service performance for a period of not less than five (5) years.
- (B) Single-Source Responsibility: Provide primers and undercoat paint produced by the same manufacturer as the finish coats.
- (C) The following publications shall become a part of this specification to the extent referenced: American Society for Testing and Materials (ASTM):
 - D3359-08 Standard Test Methods for Measuring Adhesion by Tape Test
 - E84-08 Standard Test Method for Surface Burning Characteristics of Building Materials
 - E119-08a Standard Test Methods for Fire Tests of Building Construction and MaterialsNational Fire Protection Association (NFPA):
 - NFPA 255 Standard Method of Test of Surface Burning Characteristics of Building Materials, 2006 Edition

1.04 SUBMITTALS

- (A) Before any color selections are made by the Architect, submit manufacturer's literature, indicating brand names, kind, color, texture, composition of vehicle and pigment, Federal Specification Number where applicable, and certificates as specified. List each material to be painted and cross-reference the specific-coating, finish system, and application. Items and surfaces
- (B) All paint colors to be as selected from a complete color chip catalog submitted by the general contractor and listed on color schedule, to be issued by the Architect. Do not proceed without approved schedule.

1.05 PRODUCT HANDLING

- (A) All materials shall be delivered to the building in manufacturer's sealed cans, or packages, with labels intact and seals unbroken.
- (B) The contractor shall be assigned a definite place for the storage of his equipment and the mixing of his materials and he shall have such place clean and in order at all times.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- (A) Manufacturers considered acceptable suppliers of surface preparation products specified in this section shall include and be limited to the following:
 - Pittsburg Paints/PPG Industries www.pittsburghpaints.com
 - Sherwin-Williams Company (SW) www.sherwin.com

- (B) Manufacturers considered acceptable suppliers of finished coat specified in this section shall include and be limited to the following:
 - Pittsburg Paints/PPG Industries www.pittsburghpaints.com
 - Sherwin-Williams Company (SW) www.sherwin.com
- (C) Paints of these manufacturers shall be of best grades recommended by manufacturers for the purposes intended. Oils, thinners, driers shall be only those recommended by manufacturer of paint. Nomenclature in some areas of this section is based on products and terminology of Porter Paints.
- (D) Shellac shall be fresh, pure, best grade dewaxed, white shellac, three-pound cut.

2.02 PRODUCTS SELECTIONS

- (A) Proprietary Names: The materials and schedules include in this specification are based on the products of approved manufacturers.
- (B) Approved Substitutions: Use of manufacturer's proprietary product names to designate colors or materials is not intended to imply that products named are required to be used to the exclusion of equivalent products of other manufacturers. Other paint manufacturers may be submitted to the Architect as an approved substitution. To obtain consideration as an approved substitution, such submittals shall clearly indicate the specification paragraph and item number for which the substitution is being proposed. All items being submitted for approval shall be submitted 10 days prior to the project bid in keeping with requirements of section *Substitutions*.

2.03 MATERIALS

- (A) METAL DOOR CASINGS AND DOORS PAINTING
SCHEDULE:
 - 1. Sand heavily with rough sand paper.
 - 2. Surface prep with Power Tool Cleaning PPG 90-912 or 4160 Devguard
 - 3. Pre-primer coat to be Pro-Cryl Universal Primer
 - 4. Prime coat to be Pro-Cryl Universal Primer
 - 5. Intermediate coat to be Waterbased Industrial Enamel
 - 6. Finish coat to be Waterbased Industrial Enamel
 - 7. 3 coats minimum
- (B) EXPOSED ROOF DECK, BAR JOIST AND HVAC DUCTWORK AT GYM AREA:
 - 1. Clean all surfaces to be free from dust and dirt.
 - 2. Sand areas as needed for bonding or use paint bonding primer prior to finish coats.
 - 3. If already painted and adhering well, etch the existing paint for adhesion or as recommended by the paint manufacturer.

4. Provide primer coat as required by manufacturer
5. Paint minimum of 2 coats of Sherwin-Williams Marine Grade Flat Waterborne Acrylic Dryfall paint. Color shall be white.

PART 3 EXECUTION

3.01 EXAMINATION

- (A) Store all materials at the site at least 24 hours before using in order to bring their temperature between 65 and 85 degrees F.
- (B) Do no exterior or interior painting in foggy, damp or rainy weather. When building is enclosed, interior work may be painted.
- (C) Paint exterior and interior surfaces when the ambient temperature is between 45 and 90 degrees F, except when water thinned paints are used, the ambient temperature shall be between 50 and 90 degrees F, unless otherwise designated in the manufacturer's printed instructions. Maintain these temperatures until the paint dries hard.
- (D) Examine substrates and conditions under which painting will be performed for compliance with paint application requirements. Surfaces receiving paint must be thoroughly dry before paint is applied.
- (E) Do not begin to apply paint until unsatisfactory conditions have been corrected. Start of painting will be construed as the Applicator's acceptance of surfaces and conditions within a particular area.
- (F) Coordination of Work: Review other Sections in which primers are provided to ensure compatibility of the total system for various substrates. On request, furnish information on characteristics of finish materials to ensure use of compatible primers.
- (G) Notify the Architect about anticipated problems using the materials specified over substrates primed by others.

3.02 PREPARATION

- (A) General: Remove hardware and hardware accessories, plates, machined surfaces, lighting fixtures, and similar items already installed that are not to be painted, or provide surface-applied protection prior to surface preparation and painting. Remove these items, if necessary, to completely paint the items and adjacent surfaces. Following completion of painting operations in each space or area, have items reinstalled by workers skilled in the trades involved.
- (B) For fire rated doors required to be field finished, the identifying labels are to be masked in place and painted over, not removed.
- (C) Cleaning: Before applying paint or other surface treatments, clean the substrates of substances that could impair the bond of the various coatings. Remove oil and grease prior to cleaning. Schedule cleaning and painting so dust and other contaminants from the cleaning process will not fall on wet, newly painted surfaces.

- (D) Surface Preparation: Clean and prepare surfaces to be painted according to the manufacturer's instructions for each particular substrate condition and as specified. Provide barrier coats over incompatible primers or remove and reprime. Notify Contracting Officer in writing about anticipated problems using the specified finish-coat material with substrates primed by others.
- (E) Cementitious Materials:
1. Prepare concrete, concrete masonry block, cement plaster & mineral-fiber-reinforced cement panel surfaces to be painted. Remove efflorescence, chalk, dust, dirt, grease, oils, and release agents. Roughen, as required, to remove glaze. If hardeners or sealers have been used to improve curing, use mechanical methods of surface preparation. Use abrasive blast-cleaning methods if recommended by the paint manufacturer. Determine alkalinity and moisture content of surfaces by performing appropriate tests. If surfaces are sufficiently alkaline to cause the finish paint to blister and burn, correct this condition before application.
 2. Concrete floors to be painted shall be neutralized by washing with a solution of three pounds of zinc sulphate crystals to one gallon of water, allowed to dry three days and brushed thoroughly free of crystals.
 3. Do not paint surfaces where moisture content exceeds that permitted in manufacturer's printed directions.
 4. Concrete shall have all broken and spalled edges repaired with patching compound to match adjacent surfaces. Remove projections to level of adjacent surface by grinding or similar methods.
- (F) Masonry: Remove all loose mortar in masonry work. Replace mortar and fill all open joints, holes, cracks and depressions with patching compound, finished flush with adjacent surface, with texture to match texture of adjacent surface.
- (G) Ferrous Metals: Clean ungalvanized ferrous metal surfaces that have not been shop-coated; remove oil, grease, dirt, loose mill scale, and other foreign substances. Use solvent or mechanical cleaning methods that comply with recommendations of the Steel Structures Painting Council (SSPC). Touch up bare areas and shop-applied prime coats that have been damaged. Wire-brush, clean with solvents recommended by the paint manufacturer, and touch up with the same primer as the shop coat.
- (H) Galvanized Surfaces: clean galvanized surfaces with nonpetroleum-based solvents so that the surface is free of oil and surface contaminants. Remove pretreatment from galvanized sheet metal fabricated from coil stock by mechanical methods.
- (I) Gypsum Plaster and Drywall: Remove efflorescence, loose and chalking plaster. Remove dust, dirt, and other deterrents to paint adhesion. Fill holes, cracks, and other depressions with patching compound, finished flush with adjacent surface, with texture to match texture of adjacent surface.

3.03 MIXING

- (A) Carefully mix and prepare paint materials according to manufacturer's directions.

- (B) Maintain containers used in mixing and applying paint in a clean condition, free of foreign materials and residue.
- (C) Stir material before application to produce a mixture of uniform density; stir as required during application. Do not stir surface film into material. Remove film and, if necessary, strain material before using.
- (D) Use only thinners approved by the paint manufacturer and only within recommended limits.
- (E) Tinting: Tint each undercoat a lighter shade to facilitate identification of each coat where multiple coats of the same material are applied. Tint undercoats to match the color of the finish coat, but provide sufficient differences in shade of undercoats to distinguish each separate coat.

3.04 APPLICATION

- (A) Apply paint according to manufacturer's directions. Use applicators and techniques best suited for substrate and type of material being applied. Do not paint over dirt, rust, scale, grease, moisture, scuffed surfaces, or conditions detrimental to formation of a durable paint film.
- (B) Paint colors, surface treatments, and finishes are indicated in the schedules.
- (C) Provide finish coats that are compatible with primers used.
- (D) The number of coats and the film thickness required are the same regardless of the application method. Do not apply succeeding coats until the previous coat has cured as recommended by the manufacturer. Sand between applications where sanding is required to produce a smooth even surface according to the manufacturer's directions.
- (E) Paint surfaces behind movable equipment and furniture the same as similar exposed surfaces. Before the final installation of equipment, paint surfaces behind permanently fixed equipment or furniture with prime coat only.
- (F) Paint back sides of access panels and removable or hinged covers to match exposed surfaces.
- (G) Finish interior of millwork cabinets and similar field-finished casework to match exterior.
- (H) Sand lightly between each succeeding enamel coat.
- (I) Omit primer on metal surfaces that have been shop-primed and touch-up painted.
- (J) Scheduling Painting: Apply first coat to surfaces that have been cleaned, pretreated, or otherwise prepared for painting as soon as practicable after preparation and before subsequent surface deterioration.
- (K) Allow sufficient time between successive coats to permit proper drying. Do not recoat until paint has dried to where it feels firm, does not deform or feel sticky under moderate thumb pressure, and where application of another coat of paint does not cause the undercoat to lift or lose adhesion.
- (L) Application Procedures: Apply paints and coatings by brush, roller, spray, or other applicators according to the manufacturer's directions.
 - 1. Use brushes best suited for the material applied.

2. Use rollers of carpet, velvet back, or high-pile sheep's wool as recommended by the manufacturer for the material and texture required.
 3. Use airless spray equipment with orifice size as recommended by the manufacturer for the material and texture required.
- (M) Minimum Coating Thickness: Apply materials no thinner than the manufacturers recommended spreading rate. Provide the total dry film thickness of the entire system as recommended by the manufacturer.
- (N) Mechanical and Electrical Work: Painting mechanical and electrical work is limited to items exposed in mechanical equipment rooms and in occupied spaces.
- (O) Fillers and Preparation:
1. Apply block fillers to concrete masonry block at a rate to ensure complete coverage with pores filled.
 2. Knots, sap, and pitch streaks in lumber that will be given a paint finish shall be brush coated with Shellac before the prime coat is applied.
- (P) Prime Coats:
1. Before applying finish coats, apply a prime coat of material, as recommended by the manufacturer, to material that is required to be painted or finishes and that has not been prime-coated by others.
 2. Recoat primed and sealed surfaces where evidence of suction spots or unsealed areas in first coat appears, to ensure a finish coat with not burn-through or other defects due to insufficient sealing.
 3. After the prime coat has been applied, nail holes shall be filled with putty colored to match the finish. Putty shall be brought flush with the surface of the woodwork.
- (Q) Pigmented (Opaque) Finishes:
1. Completely cover to provide a smooth, opaque surface of uniform finish, color, appearance, and coverage.
- (R) Cloudiness, spotting, holidays, laps, brush marks, runs, sags, ropiness, or other surface imperfections will not be acceptable.
- (S) Provide satin finish for final coats.
- (T) Paint shall be applied in the number of coats specified, which is minimum acceptable and at the square foot coverage as stated in the paint manufacturer's printed specifications. It is intended that paint so applied shall cover to the satisfaction of the architect or additional coats shall be applied at the contractor's expense until approval is obtained.
- (U) Completed Work shall match approved samples for color, texture, and coverage. Remove, refinish, or repaint work not complying with specified requirements.
- (V) The painting contractor shall be responsible for marking the necessary identifications of rated partitions above the finish ceiling as required by the International Building Code: "Corridor partitions, smokestop partitions, horizontal exit partitions, exit enclosures, and fire walls shall be effectively and permanently identified with signs or stenciling in a manner acceptable to the authority having jurisdiction. Such identification shall be above any decorative ceiling and in concealed spaces." Wording shall be "FIRE AND SMOKE BARRIER – PROTECT ALL OPENINGS".

3.05 FIELD QUALITY CONTROL

- (A) The Owner reserves the right to invoke the following test procedure at any time and as often as the Owner deems necessary during the period when paint is being applied:
1. The Owner will engage the services of an independent testing agency to sample the paint material being used. Samples of material delivered to the Project will be taken, identified, sealed, and certified in the presence of the Contractor.
 2. The testing agency will perform appropriate tests for confirmation of the product compliance with these specifications.
 3. If test results show material being used does not comply with specified requirements, the Contractor may be directed to stop painting, remove noncomplying paint, pay for testing, repaint surfaces coated with rejected paint, and remove rejected paint from previously painted surfaces if, upon repainting with specified paint, the two coatings are compatible.

3.06 CLEANING AND PROTECTION

- (A) The contractor shall take particular care by use of clean drop cloths, masking and other suitable means to protect adjoining surfaces, fixtures, and materials of all kinds, and shall be held responsible for and shall repair any damage resulting from the painting operations.
- (B) At the end of each workday, remove empty cans, rags, rubbish, and other discarded paint materials from the site.
- (C) After completing painting, clean glass and paint-spattered surfaces. Remove spattered paint by washing and scraping. Be careful not to scratch or damage adjacent finished surfaces.
- (D) Protect work of other trades, whether being painted or not, against damage by painting. Correct damage by cleaning, repairing or replacing, and repainting, as acceptable to Architect.
- (E) Provide "Wet Paint" signs to protect newly painted finishes. Remove temporary protective wrappings provided by others to protect their work after completing painting operations.
- (F) At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

END OF SECTION

SECTION 10 74 49 - COURT EDGE REDUCER RAMP

PART ONE - GENERAL

1.01 SYSTEM DESCRIPTION

These specifications are for the CourtEdge™ Reducer Ramp products selected and dimensioned to provide the transition piece from existing flooring or changes to the substratum (Ramps modify existing changes in vertical rises). CourtEdge™ Reducer Ramps can be applied surfaces including but not limited to concrete, tile, VCT, marble, wood, asphalt, natural and synthetic fibers and painted surfaces.

1.02 QUALITY ASSURANCE

A. Supplier Qualifications

Supplier of this Ramp product shall be SafePath™ Products by Van Duerr Industries, Inc., 21 Valley Court, Chico, CA 95973, PH# (530) 893-1596, (800) 497-2003, Fax (530) 893-1560, www.SafePathProducts.com or a registered supplier of Van Duerr Industries, Inc. or SafePath™ Products.

Installer or Contractor Qualifications: Installations will be performed by skilled tradesman experienced and trained in the flooring modifications and alterations.

1.03 REFERENCE STANDARDS

- A. Manufacturer provides each ramp in accordance with the model descriptions and specifications established herein. All ramps shall be installed in accordance with all governing agencies with jurisdiction and in accordance with the Americans with Disabilities Act (ADA, 1990). CourtEdge™ Reducer Ramp products shall be in compliance with the ADA and Accessibility Guidelines (ADAAG) Sections 303 (Changes in Level) and 405 (Slope & Rise) of the Federal Register, 28 CFR, Part 36 and ANSI Standard A117.1. (1992), CABO and California Title 24, unless otherwise specified by the customer. All ramps and reducers shall exceed 8.33% maximum slope requirement.

1.04 PERFORMANCE

- A. CourtEdge™ Reducer Ramp products meet or exceed the following tests:

ASTM D 412 Tensile Elasticity 630psi

ASTM D395-82 Comp. Test 7.84%

ASTM D 2240 Hardness 65 (Shore A)

ASTM C 1026-84H20 Absorb. .75%

Density 1.01 grams/cm³

- 1) CourtEdge™ Reducer Ramp products have no load weight limitations when properly installed according to manufacturer instructions.

CourtEdge™ Reducer Ramps exceeded all recognized standards utilizing three test methodologies for Static Coefficient of Friction (SCOF), Fixed-Angle Ramp-Walking, and Surface Roughness.

1) Static Coefficient of Friction (SCOF)

ASTM C 1028-96 (Dry- 0.95)

ASTM C 1028-96 (Wet- 0.79)

Static Coefficient of Friction values of 0.60 or higher are considered adequate for level floors in normal pedestrian traffic in commercial areas. Sliders used in testing were of Neolite synthetic rubber. Other materials used for SCOF included:

ASTM C 1028-96 Dry Leather (0.79)

ASTM C 1028-96 Wet -4 S rubber (0.82)

ASTM C 1028-96 Wet TRL rubber (0.95)

ASTM F609 Rubber 0.80

2) Fixed Angle Walking

Utilizing two test subjects, one male and one female, facing a down slope and an up slope at a flooring angle of 27 degrees with wet running water. Angle of degree corresponded to a SCOF of 0.50. Under normal walking conditions a SCOF of 0.40 (22 degrees) is a widely accepted minimum for walking tests of flooring for level surfaces. Testing angle appreciably exceeds the minimum safety criterion.

- a. bare foot
- b. Red Wing Iris athletic shoes
- c. Converse 2D 1292 tennis shoes
- d. Naturalizer 78ON73 wedge sandals

3) Surface Roughness – surface total-mean peak-to-valley roughness, R_{tm} or R_Z, measured by a Surtronic 10 profilometer was 25. Water-wet surface is low when the roughness exceeds 20 microns.

1.05 DELIVERY, STORAGE and HANDLING

- A. Store products in accordance with the manufacturer's instructions, with labels intact. CourtEdge™ Reducer Ramps shall be kept dry and away from sources of heat. Store on flat level surfaces.

1.06 PROJECT CONDITIONS

- A. Review installation procedures and coordinate CourtEdge™ Reducer Ramp installation with other work.
- B. Do not install on frozen materials or materials mixed or coated with ice or frost.
- C. Surface must be dry and clean and free from dirt, oil, frost and all other foreign materials and have a minimum surface temperature of 35°F (3°C)

PART TWO - PRODUCTS

2.01 MANUFACTURER

- A. CourtEdge™ Reducer Ramp products shall be manufactured by SafePath™ Products by Van Duerr Industries, Inc., 21 Valley Court, Chico, CA 95973, PH# (530) 893-1596, (800) 497-2003, Fax (530) 893-1560, www.SafePathProducts.com

2.02 MATERIALS

- A. CourtEdge™ Reducer Ramp material shall be made from 100% reclaimed 20 mesh minus crumb rubber with urethane binders by compression molding, and have a maximum hardness (Shore A) of 65 and comply with Americans with Disabilities Act (1990) ADAAG 4.2.5
- B. Changes in level and ADAAG 4.8.2 Slope and Rise, UFAS Unit 4, Thresholds and ANSI Standards 117.1 (1986). Ramps shall have a 3/16" (6.35mm) front radius maintaining a slope of less than 1:12 to the top of the Ramp.
- C. Ramp surfaces for models in 1150 through 2395 series shall have molded slip resistant traction pads 2" (50.8mm) by 3 3/8" (85.725mm) with 1/8" (3.175mm) depressed channel.
- D. Ramp surfaces for models in Reducer 1400, 1200, 1000 & 900 series shall have molded slip resistant traction diamond design with equal and proportional spacing.
- E. Ramps shall come in standard heights of single or multi piece construction in accordance with the model description and specifications established herein.

Adhesive: WARNING: NEVER USE CONSTRUCTION ADHESIVE i.e. LIQUID NAILS™ OR OTHER SIMILAR NON-APPROVED COMPOUNDS TO FASTEN RAMPS

- F. Securing ramp product to substratum. One part non-corrosive silicon adhesive sealant shall be one of the following products:
 - 1) SIKAFLEX 1A – One part Polyurethane, Elastomeric sealing/adhesive. Meeting Federal specifications TT-S-00230C, Type II, Class A. Meets c-920, Type S, Grade NS, use T, NT, O, M, G, I; Canadian standard CAN/CGSB 19.13-M87.
 - 2) SU 5007 RTV SILICONE ADHESIVE SEALANT. A supplied Silicones Unlimited Inc. located at 4577 Blakedale Circle, Roswell, GA 30075, PH# (770) 643-1880, or supplied by their authorized distributors. Silicone adhesive shall have Tensile Strength ASTM D-412 150 psi. UV Resistant, Curing time @ 77°F and 50% Relative Humidity (1/8" thickness) of 24 hour. Hardness (Shore A) 28. Tack-Free time @ 77°F and 50% Relative Humidity, 20-30 minutes, and temperature range -80°F (62°C) to 400°F (205°C) or one part DOW CORNING 995 SILICONE STRUCTURAL ADHESIVE. Supplied by Dow Corning Corporation, Midland, MI 48686, PH# (517) 496-6000 or one of their authorized suppliers. Silicone adhesive shall have Tensile Strength ASTM D-412 350 psi. UV Resistant, Curing time @ 77°F and 50% Relative Humidity (1/8" thickness) of 24 hour. Hardness (Shore A) 28. Tack-Free time @ 77°F and 50% Relative Humidity, 1.5 hours. Adhesive temperature range -65°F (-54°C) to 350°F (-65°C).

2.03 COATING MATERIALS

SafePath™ Products' StoneCap™ coating system:

1. StoneCap™ coating is a two-component, rapid-curing, hand-applied, non-solvent protective coat that can be trafficked within 4 hours of application at 68°F.
2. StoneCap™ is broadcast to rejection with the chosen dry skid resistant aggregate as approved by the manufacturer. This material is broadcast by manufacturer and applied to provide for even surface and 100% coverage of substrate material. Typically the DFT of the traffic coat including aggregate shall be 80 mils.
3. The StoneCap™ topcoat is a one-component, rapid-curing, clear, non-solvent polyurethane sealer which can be trafficked in 4 hours at 68°F.
4. StoneCap™ coating is applied by the manufacturer and completely cured prior to shipment of product.

PART THREE - EXECUTION

Ramps shall be installed in accordance with manufacturer's instructions using ONLY recommended non-corrosive silicone adhesive sealants or mechanical fasteners.

3.01 PREPARATION

- A. Surfaces must be dry, clean, and free from dirt, oil, frost, and all other foreign materials, and have a minimum surface temperature of 35°F (3°C). CourtEdge™ Reducer Ramps can be applied over expansion joints up to ½" (12mm) in width. In accordance with manufacturer's guidelines, fill holes and cracks greater than ½" (12mm) in depth. Remove all bumps and foreign surfaces. Abrade highly polished and waxed surfaces.
- B. Field verify all dimensions for vertical heights and path widths.

END OF SECTION

DIVISION NO. 11 – EQUIPMENT

SECTION 11 66 00 - SPECIALTY EQUIPMENT - ATHLETICS

PART 1 GENERAL

1.01 REQUIREMENTS: The requirements of all sections of Division One apply to work under this section.

1.02 WORK INCLUDED: Furnish labor and materials to complete items indicated, specified or both.

PART 2 PRODUCTS AND EXECUTION

2.01 VOLLEYBALL SLEEVE INSERTS

(A) Provide two (2) volleyball inserts, 00870-200, 3-1/2" diameter floor sleeve by Porter Athletic Equipment. Sleeves shall be installed flush with gym floors as recommended by manufacturer. Sleeves shall come with brass covers.

END OF SECTION

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DIVISION NO. 12 – FURNISHINGS

SECTION 12 66 13 - TELESCOPING GYMNASIUM BLEACHERS

PART 1 GENERAL

1.01 WORK: The requirements of all sections of Division One apply to work under this section.

- (A) Telescoping gymnasium bleachers shall be purchased and installed by contractor. Interkal – basis of design. Approved equals will be reviewed if all aspects of specifications are met.

1.02 RELATED WORK:

- (A) Electrical section.
(B) Gymnasium flooring; Section 09 64 66.

1.03 REFERENCES: Applicable building codes IBC Edition Year 2021

1.04 DESCRIPTION OF THE SYSTEM:

- (A) The bleacher system shall be comprised of multiple tiered, closed deck seating rows operating in a telescopic manner, incorporating the most economical quantity of sections while still complying with all loading requirements.
(B) The first moving row shall be secured with friction or mechanical locks. Other rows shall be mechanically locked, operable only upon unlocking and cycling the first row, quantity to be determined by Interkal engineering.
(C) Each bleacher row shall be comprised of risers, seat and deck components, and a complete set of supportive columns and braces.
(D) The telescopic bleacher shall incorporate a locking system permitting the use of one, several, or all rows, each locked in the extended position.

1.05 QUALITY ASSURANCE

- (A) Qualifications
1. Manufacturing: Manufacturer shall be regularly engaged in the design and manufacturing of telescopic seating for not less than twenty years.
2. Engineering: It will be mandatory that each bidder submit with their bid an affidavit signed by a Registered Professional Engineer stating that the product to be supplied has been tested by an independent testing facility and meets all applicable code requirements.
(B) Deviations: It will be the responsibility of the bidder to furnish with their bid, a list clarifying any deviations from the specifications, written or implied. Those bidders not submitting a list of deviations will be presumed to have bid as specified.
(C) Guarantees:
1. One-Year Guarantee: The manufacturer shall guarantee all work performed under these specifications to be free from defects for a period of one year.
(D) Product Improvements: Seating provided shall incorporate manufacturer's design improvements and materials current at time of shipment.

1.06 SUBMITTALS:

- (A) Submit manufacturer's installation instructions and descriptive literature in accordance with Division 01.
- (B) Manufacturer's operating and maintenance manuals in accordance with Division 01.

1.07 DESIGN CRITERIA:

- (A) Telescopic bleacher design and fabrication shall conform to IBC 2021 and ANSI A117.
- (B) Telescopic gymnasium seating will be designed to support a vertical live load of 100 PSF, but not less than 120 PLF on both seat boards and footboards. Seating shall also be designed to carry a horizontal sway force of 24 PLF parallel to the seating and 10 PLF perpendicular to the seating.
- (C) Steel components shall be cold-formed from appropriate width strip stock conforming to ASTM A570 - Grade C 30KSI, ASTM A653- Grade 33 and 50, ASTM A500 - Grade B 46 KSI as applicable.
- (D) Lumber components are kiln dried, finger jointed, edge glued southern pine of grade "B & B Finish" manufactured to the current SPIB glued-laminated standards for southern pine.
- (E) Plywood deck boards shall be fabricated from Douglas Fir Premium Underlayment with exterior glue, 5 ply minimum, solid crossband directly under face ply, species Group 1 and manufactured in accordance with PS-1-95.

PART 2 PRODUCTS

2.01 MANUFACTURER:

- (A) Telescopic seating as manufactured by Interkal, Kalamazoo, Michigan, is the standard of quality required and specified herein (or approved equal). Approved Equals: Hussey, Irwin Seating.

2.02 MATERIALS:

- (A) Model: Interkal, closed deck telescopic bleachers
- (B) Type: Wall attached
- (C) Quantity:
 - 1. Provide 2 banks of wall-attached 12 rows high.
- (D) ADA
 - 1. Notchouts: Provide a 36" wide wheel chair space as shown on the plans and as required to meet local code jurisdiction compliance with ADA. (One row deep).
 - 2. Truncations: Provide a full section truncation with all necessary front rails, closure panels, and portable step assemblies at aisles as required to meet local jurisdiction compliance with ADA.
- (E) Dimensions:
 - 1. Rise per row: 10 -1/4"
 - 2. Row to row spacing 22", 24" (See plans).
- (F) Propulsion
 - 1. Friction Power- Furnish Interkal friction power, integral automatic electro-mechanical propulsion system to open and close telescopic seating system. Operation shall assure full visual control of the seating bank. The Wide Track System incorporates two friction drive roller assemblies as an integral part of both first row vertical column assemblies. Each section of bleacher shall have a power system that shall consist of two vertical column roller assemblies which shall include two 6" diameter by 2 1/2" wide cast drive wheels for a minimum of four friction roller contact points per section of bleacher. Each

roller shall have a specially formulated 45-durometer rubber covering to grip the floor as the units roll in and out. The two friction drive roller assemblies shall be installed a minimum of 7-feet apart per section. The two friction roller assemblies are linked together by a continuous drive shaft driven by a 1/2 H.P. 208V, 3-phase motor that shall enable the rollers to work simultaneously, resulting in a more efficient operation with allowance for minor variations in the floor surface. All floor friction power systems shall be controlled by a dual directional, removable walk along pendant which plugs into the front of the first row to give the operator proper position for visual control. The pendant control voltage shall be 24 VAC @ less than 50 MA for the safety of all operating personnel. **The entire power system shall be U.L. Recognized.** A 208/220 volt 3-phase power source, including conduit, wiring, and safety disconnect must be provided by others. The electrical contractor shall perform the connections to the seating equipment at the safety disconnect. Motors, housing, and wiring shall be installed by certified personnel. (VERIFY EXISTING ELECTRICAL POWER IN FIELD PRIOR TO ORDERING MATERIAL)

2.03 ACCESSORIES:

- (A) Foot Level Aisles: Provide footrest level aisles at locations and sizes as shown on plans and approved shop drawings.
 - 1. Center Aisle: Provide a permanently attached self-storing aisle rail, which is designed to eliminate all labor associated with set up and storage of the aisle rails.
 - 2. Intermediate Steps: Provide manufacturers standard intermediate step as necessary per applicable code.
- (B) Last Row Closure
 - 1. Rear Closure Board: Provide and install a properly supported, flush mounted board between the last row of the bleacher and the wall.
- (C) Wheelchair Seating:
 - 1. Notchouts: Provide manufacturers standard permanent handicap notchout (36" wide) located as shown on architectural plans. Notchouts must be located at section joints only to avoid interference with understructure. Fascia panels shall have manufacturers standard polydeck finish to match deck board surface. Available in one row or two row deep, (One row deep).
- (D) Front Railing: Provide rigid 36" high, fixed tubular steel rail with vertical intermediate members to fill design criteria. Rail to be mounted full width at all two row deep ADA wheelchair accommodations. Finish shall be a polyester powder coat. Front rails are to be designed to comply with all applicable codes and remain consistent with all other rails not allow clearance of a 4" sphere.
- (E) End Railing:
 - 1. Self-Storing End Rails: Provide steel self-storing 42" high self-storing end guard rails with tubular supports and vertical intermediate members to comply with all code requirements. Rails shall be fitted to each exposed bank end from third row and above with all steel to steel connections. Finish shall be a polyester powder coat.
- (F) Aisle Railing: Shall be self storing.
- (G) Operation:
 - 1. Pendant Control: Provide pendant control style operation for the bleachers. Extension and retraction shall be accomplished by use of the pendant control plugged into a single receptacle. The receptacle shall be mounted at the first row.
- (H) Numbering: Provide seat numbers and row letters for sculpture seat modules. Sequence to be determined by architect or owner.

- (I) Back Panels: (For reverse fold and mobile units) Provide the manufacturers standard polydeck finish to match deck board surface. Back panels will be provided a maximum of 8' high.
- (J) Back Rails: (For reverse fold and mobile units) Provide the manufacturers standard back rails with vertical intermediate members to eliminate ladder effect and comply to all applicable building codes. Back rails are to be designed to not allow clearance of a 4" sphere.
- (K) Vinyl-End Curtains: Provide manufacturers standard vinyl end curtains to close off under the bleacher units in the extended position. Curtain color is to be selected from manufacturers standard offering.
- (L) End Panels: Provide manufacturers standard end panels to close off the opening between end rails and the wall when the bleachers are stacked. (Not available with vinyl end curtains)

2.04 FABRICATION

- (A) Continuous Wheel Channel: Wheel channels shall consist of a one piece formed steel channel welded to the base of a vertical column. Wheel channels accommodate 8 to 12 wheels per row for maximum weight distribution and operating ease. The number of wheels increase as the number of rows increase.
- (B) Wheels: 3-1/2" diameter with 1-1/8" non-marring soft rubber face with rounded edges designed to protect wood or synthetic floor. Provide 1/2" diameter axle for all wheels
- (C) Columns: Electrically welded closed rectangular steel tube, 2" x 3" minimum size, 14 gauge steel fitted with a rear welded gusset at the wheel channel.
- (D) Row Interlocks: Join each row structure front to rear by means of two (2) interacting steel connections, plus automatic gravity row locks where Engineering determines they are required.
 - 1. Lower: Lower track guides shall be an external superslide rod to guarantee positive engagement of vertical supports without binding and assures smooth operation over uneven floor conditions.
 - 2. Upper: Upper track guides shall completely interlock adjacent understructure support. A welded stop to ensure correct extension of bleacher unit on deck support. Use of bolt and nut stops are not acceptable, due to risk of loosening.
- (E) Diagonal Braces: Structural formed steel truss fitted to rows 4 and beyond. Bracing shall be attached to the rear riser at optimum locations to insure structural integrity. Bracing will be designed and shaped to support a minimum load of 1000(lbs) of both compression and tension forces created when the bleacher is loaded.
- (F) Deck Supports: Shall be of structural steel, 11 gauge spaced not greater than 60" on center for maximum deck stiffness.
 - 1. Rollers: Every deck support not attached to a vertical post will have an integral nylon roller to avoid steel to steel friction points for more efficient operation.
- (G) Decking: All deck boards shall consist of 19/32" nominal Douglas Fir CC grade plywood with exterior glue and solid crossbands. An extruded aluminum "H" connector shall be placed between plywood panels. Exposed wear surfaces shall be finished with a layer of high Density polyethylene plastic .025 - .030 thick, Light Gray in color, complimentary to the seat option. Deck finishes, such as clear coat, requiring more than simple touch up to restore it to a new appearance after wear occurs are unacceptable.
- (H) Welds: All welds shall be made at the factory by welders that are AWS certified on the equipment and process used.
- (I) Nose Beam: Shall be one-piece 13-gauge galvanized steel. 13-gauge steel is utilized for the necessary structural integrity to accommodate section lengths up to 26'

- (J) Rear Riser: Shall be one piece formed 14-gauge, grade 50, galvanized steel, with a continuous access joint to fully encapsulate footrest panel for ease of cleaning and additional structural support. 14-gauge roll formed steel is utilized for the necessary structural integrity to accommodate section lengths up to 26'.
- (K) Splice Plates: (For Friction or Non-Friction power only) Each section joint shall be tied together with two structural steel members per row, employing a minimum of four steel to steel through bolt connections at the nose beam and a minimum of eight steel to steel through bolt connections at the lower steel rear riser. Gauge of splice plates to match the gauge of the nose beam and rear riser. Splice plates employing steel to plywood deck board attachments will not be acceptable. Gauge of splice plates to match the gauge of the nose beam and rear riser. In order to minimize deflections and keep rows in alignment during operation, splice connections shall transfer both axial loads (tension/compression) and bending.
- (L) Fasteners: All structural connections shall be made with S.A.E. grade 5 or better stress rated bolts. The use of self-tapping bolts is not acceptable.
- (M) Finish:
 1. Steel Understructure abraded, cleaned and finished with russet brown water base acrylic paint. Steel risers and nose beams finished with corrosion resistant silver gray matte finish with galvanized alloy plating.

2.05 SEAT OPTIONS:

- (A) SculptureSeat Modules:
 1. 18-inch wide one-piece individual seating modules shall be constructed of high-density polyethylene. Provide in 10" deep.
 2. Each module shall have two longitudinal and five transverse internal ribs to provide additional structural integrity and resistance to impact.
 3. Each module shall have a full ½" interlock to the adjacent module both around the perimeter and along the internal ribs to eliminate pinching hazards and assures proper alignment.
 4. A steel-to-steel attachment of each module to a minimum 14 gauge galvanized steel nosebeam shall be provided for maximum rigidity. All such mounting hardware shall be concealed.
 5. End caps shall be provided at the ends of each bank (section, if manual) of seating as well as at each aisle.
 6. Each module shall have a recessed area for optional seat numbering.
 7. Select from manufacturers 15 standard solid colors.

PART 3 EXECUTION

3.01 INSPECTION

- (A) Verify that areas to receive telescopic bleachers are free from impediments interfering with installation.
- (B) Do not begin work until building conditions are satisfactory.

3.02 INSTALLATION

- (A) Install telescopic bleachers in accordance with manufacturer's instructions and approved submittal drawings.
- (B) Adjust bleachers for smooth and proper operation.
- (C) Clean bleachers and remove all debris from gymnasium resulting from installation.

END OF SECTION

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